Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-M-Q1 · question page 425

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

Which statement is a TAUTOLOGY (true in every row of its truth table)?

Answer: B. (P and Q) or not P or not Q

Reasoning: If P and Q are both true, the first bracket is true; otherwise "not P or not Q" is true. So it is always true.

- ✗ P or Q: false when both are false
- ✗ if P then Q: false when P is true and Q false
- ✗ P and Q and not Q: false in every row

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-M-Q2 · question page 426

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

Which statement is a CONTRADICTION (false in every row)?

Answer: D. P and Q and not Q

Reasoning: Q and not Q can never both be true.

- ✗ P or not P: a tautology
- ✗ if P then Q: a contingency
- ✗ P or Q: a contingency

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-M-Q3 · question page 427

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

Complete the truth table for "(if P then Q) or (if Q then P)".

Answer: T T T / F T T / T F T / T T T

Reasoning: In every row at least one of the two conditionals is true, so the last column is all T: a tautology.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S19-M-Q4 · question page 428

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

"not (P and Q)" is logically equivalent to "(not P) or (not Q)".

Answer: True

Reasoning: Both are false only when P and Q are both true (De Morgan's law).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-M-Q5 · question page 429

LOGIC.TRUTH.02 · Classify statements as tautologies, contradictions or contingencies, and test equivalence

How is "if P then Q" classified?

Answer: A. a contingency: true in some rows and false in one

Reasoning: It is false only in the row P true, Q false, so it is a contingency.

- ✗ a tautology: it is false when P is true and Q false
- ✗ a contradiction: it is true in three rows
- ✗ neither true nor false: every row has a value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-B1-Q1 · question page 430

LOGIC.TRUTH.01 · Fill in truth tables

Complete the truth table for "P or not P".

Answer: F T / T T

Reasoning: Always true.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S19-B1-Q2 · question page 431

LOGIC.TRUTH.01 · Fill in truth tables

Complete the truth table for "P and not P".

Answer: F F / T F

Reasoning: Always false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (2 rows).

S19-B1-Q3 · question page 432

LOGIC.TRUTH.01 · Fill in truth tables

Complete the truth table for "not P or Q".

Answer: T / F / T / T

Reasoning: False only when P true and Q false.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S19-B1-Q4 · question page 433

LOGIC.TRUTH.01 · Fill in truth tables

How many rows does a truth table for P and Q have?

Answer: 4

Reasoning: 2 x 2.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S19-B1-Q5 · question page 434

LOGIC.TRUTH.01 · Fill in truth tables

A truth table for three statements P, Q and R has 6 rows.

Answer: False

Reasoning: Each of the 3 statements can be true or false: 2 x 2 x 2 = 8 rows.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-B2-Q1 · question page 435

LOGIC.TRUTH.02 · Always, never, sometimes

A statement is true in every row of its table. What is it called?

Answer: B. a tautology

Reasoning: Always true = tautology.

- ✗ a contradiction: that is never true
- ✗ a contingency: that is sometimes true
- ✗ a premise: a role in an argument, not a class

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S19-B2-Q2 · question page 436

LOGIC.TRUTH.02 · Always, never, sometimes

A statement is false in every row. What is it called?

Answer: C. a contradiction

Reasoning: Never true = contradiction.

- ✗ a tautology: always true
- ✗ a contingency: sometimes true
- ✗ a conclusion: not a class

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S19-B2-Q3 · question page 437

LOGIC.TRUTH.02 · Always, never, sometimes

A contingency is true in every row of its truth table.

Answer: False

Reasoning: A statement true in every row is a tautology. A contingency is true in some rows and false in others.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-B2-Q4 · question page 438

LOGIC.TRUTH.02 · Always, never, sometimes

A column reads T, T, F, T. How is the statement classified?

Answer: D. a contingency

Reasoning: Mixed values: contingency.

- ✗ a tautology: it has an F
- ✗ a contradiction: it has a T
- ✗ it cannot be told: the column decides it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S19-B2-Q5 · question page 439

LOGIC.TRUTH.02 · Always, never, sometimes

"7 is odd and 7 is not odd" is true.

Answer: False

Reasoning: It has the form "P and not P", a contradiction: it is false in every case.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-F1-Q1 · question page 440

LOGIC.TRUTH.02 · Equivalences by truth table

Complete the table to compare "not (P or Q)" with "(not P) and (not Q)".

Answer: F F / F F / F F / T T

Reasoning: The columns match: De Morgan's law.

Grading: Every output cell must match (T/F, true/false or 1/0).

Verified: Automatic: every output cell computed from the formula (4 rows).

S19-F1-Q2 · question page 441

LOGIC.TRUTH.02 · Equivalences by truth table

"if P then Q" is equivalent to "if Q then P".

Answer: False

Reasoning: They differ when exactly one of P, Q is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-F1-Q3 · question page 442

LOGIC.TRUTH.02 · Equivalences by truth table

Which statement is equivalent to "not (if P then Q)"?

Answer: B. P and not Q

Reasoning: "if P then Q" is false exactly when P is true and Q false.

✗ not P and not Q: true when both are false, but the original is false there

✗ if not P then not Q: that is the inverse

✗ not P or Q: that is the conditional itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-F1-Q4 · question page 443

LOGIC.TRUTH.02 · Equivalences by truth table

How many of these are tautologies? "P or not P"; "P and not P"; "if P then Q"; "(if P then Q) or (if Q then P)"; "P iff not P".

Answer: 2

Reasoning: "P or not P" and "(if P then Q) or (if Q then P)".

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S19-F1-Q5 · question page 444

LOGIC.TRUTH.02 · Equivalences by truth table

The negation of a tautology is also a tautology.

Answer: False

Reasoning: If a statement is true in every row, its negation is false in every row.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S19-F2-Q1 · question page 445

LOGIC.TRUTH.02 · Laws of logic

Which statement is equivalent to "P or (P and Q)"?

Answer: A. P

Reasoning: If P is true, the statement is true; if P is false, both parts are false. So it equals P (absorption).

- ✗ Q: differs when P true, Q false
- ✗ P and Q: differs when P true, Q false
- ✗ P or Q: differs when P false, Q true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S19-F2-Q2 · question page 446

LOGIC.TRUTH.02 · Laws of logic

"((if P then Q) and P) implies Q", written "if ((if P then Q) and P) then Q", is a tautology.

Answer: True

Reasoning: This is modus ponens written as one statement; it is true in all four rows.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-F2-Q3 · question page 447

LOGIC.TRUTH.02 · Laws of logic

"if ((if P then Q) and Q) then P" is a tautology.

Answer: False

Reasoning: Row P false, Q true: the "if" part is true and P is false, so the statement is false (affirming the consequent).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S19-F2-Q4 · question page 448

LOGIC.TRUTH.02 · Laws of logic

Of the 16 possible truth-table columns for statements in P and Q, how many are contingencies?

Answer: 14

Reasoning: All 16 columns except all-T (tautology) and all-F (contradiction).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S19-F2-Q5 · question page 449

LOGIC.TRUTH.02 · Laws of logic

Which pair are equivalent?

Answer: D. "if P then (Q and R)" and "(if P then Q) and (if P then R)"

Reasoning: Both sides of the first pair are false exactly when P is true and at least one of Q, R is false.

- ✗ "if (P or Q) then R" and "(if P then R) or (if Q then R)": P true, Q false, R false: the first is false, the second true
- ✗ "P and (Q or R)" and "(P and Q) or R": P false, R true: the first is false, the second true
- ✗ "not (P and Q)" and "not P and not Q": P true, Q false: the first is true, the second false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-M-Q1 · question page 451

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 3 sheets: A, B, C. Put the sheets in order from the BOTTOM sheet to the TOP sheet.

Answer: sheet B → sheet C → sheet A

Reasoning: Overlaps: B is under C, C is under A, B is under A. A's outline is complete, so A is on top; then C; B is at the bottom: B, C, A.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S20-M-Q2 · question page 452

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. List the sheets starting with the one at the BOTTOM and ending with the one on TOP.

Answer: sheet B → sheet C → sheet A → sheet D

Reasoning: At each overlap the cut-off outline is underneath: B is under C, A and D; C is under A; A is under D. So the chain is B, C, A, D from bottom to top.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S20-M-Q3 · question page 453

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Four round discs are stacked with their centers lined up, from bottom to top: W (radius 5 cm), X (radius 3 cm), Y (radius 4 cm), Z (radius 1 cm). Looking straight down from above, which discs can be seen?

Answer: C. W, Y and Z

Reasoning: A disc can be seen if every disc above it is smaller. W: all above are smaller. X: Y (4 cm) is above and larger, so X is hidden. Y: only Z (1 cm) is above. Z: on top.

- ✗ W, X, Y and Z: X is entirely covered by the larger disc Y above it
- ✗ Z only: larger discs below Z show around it
- ✗ W and Z: Y shows as a ring between radius 1 and 4 cm

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-M-Q4 · question page 454

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 4 sheets: A, B, C, D. Put the sheets in order from the BOTTOM sheet to the TOP sheet.

Answer: sheet C → sheet D → sheet A → sheet B

Reasoning: Overlaps: C is under D and A; D is under A; A is under B. From bottom to top: C, D, A, B.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S20-M-Q5 · question page 455

LOGIC.SPATIAL.03 · Infer stacking order from what is visible

Rectangular cards are placed one at a time on an 8 by 8 grid, each covering a block of cells (rows and columns numbered 1 to 8; "rows 1-4, columns 1-4" means a 4 by 4 block). In order: A covers rows 1-4, columns 1-4; B covers rows 3-6, columns 3-6; C covers rows 2-5, columns 2-5; D covers rows 1-3, columns 4-7; E covers rows 5-7, columns 1-3. Seen from above, how many of the five cards can be seen

Answer: 5

Reasoning: Check each card for a cell not covered by a later card. A: cell (1,1). B: cell (6,6). C: cell (4,4). D and E are near the top of the pile and show fully or partly. All 5 can be seen.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-B1-Q1 · question page 456

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Where sheets X and Y overlap, the outline of X runs straight through and the outline of Y stops at X's edge. What follows?

Answer: C. Y is under X.

Reasoning: The sheet whose outline is cut off is underneath.

- ✗ X is under Y.: the cut-off outline is the lower one
- ✗ They are side by side.: they overlap
- ✗ It cannot be told.: the outlines decide it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S20-B1-Q2 · question page 457

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Pairs found: P under Q, Q under R, P under R. Order the sheets from bottom to top.

Answer: sheet P → sheet Q → sheet R

Reasoning: P, Q, R.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S20-B1-Q3 · question page 458

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

A sheet whose whole outline can be seen must be on top of every sheet it overlaps.

Answer: True

Reasoning: If any sheet above it overlapped it, part of its outline would be hidden.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S20-B1-Q4 · question page 459

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Discs are stacked with centers lined up, from bottom to top: radius 2 cm, then 6 cm. Which can be seen from above?

Answer: D. only the 6 cm disc

Reasoning: The 6 cm disc covers the 2 cm disc completely.

- ✗ both discs: the larger disc above covers the smaller one
- ✗ only the 2 cm disc: it is underneath
- ✗ neither disc: the top disc is visible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-B1-Q5 · question page 460

LOGIC.SPATIAL.03 · Strategy: the cut-off sheet is underneath

Five sheets are stacked. Exactly one of them is on top of all four others. How many sheets are under it?

Answer: 4

Reasoning: All four others are under the top sheet.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F1-Q1 · question page 461

LOGIC.SPATIAL.03 · Larger stacks

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. Put the five sheets in order from the BOTTOM sheet to the TOP sheet.

Answer: sheet E → sheet D → sheet A → sheet C → sheet B

Reasoning: Every pair overlaps here. B is cut off by no sheet (top); C is cut off only by B; A by C and B; D by A, C and B; E by all four. From bottom to top: E, D, A, C, B.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S20-F1-Q2 · question page 462

LOGIC.SPATIAL.03 · Larger stacks

Rectangular paper sheets are lying flat, one on top of another, seen from directly above. Each sheet is white with a dark outline and a letter in its corner. A sheet hides the parts of the sheets under it. This stack has 5 sheets: A, B, C, D, E. List the five sheets starting with the BOTTOM sheet and ending with the TOP sheet.

Answer: sheet B → sheet E → sheet A → sheet C → sheet D

Reasoning: Every pair overlaps. D is cut off by no sheet (top); C only by D; A by C and D; E by A, C and D; B by all four. From bottom to top: B, E, A, C, D.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: every stacking order was drawn on a fine grid and compared with the picture. Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S20-F1-Q3 · question page 463

LOGIC.SPATIAL.03 · Larger stacks

Cards are placed one at a time on an 8 by 8 grid: A covers rows 2-3, columns 2-3; B covers rows 1-4, columns 1-4; C covers rows 4-6, columns 4-6; D covers rows 1-2, columns 5-6; E covers rows 3-6, columns 1-4. Seen from above, how many of the five cards can be seen at least partly?

Answer: 4

Reasoning: A lies inside B and B was placed later, so A is hidden. B still shows in rows 1-2, columns 1-4. C shows in columns 5-6 (E only covers column 4 of C). D and E show. So B, C, D and E: 4 cards.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F1-Q4 · question page 464

LOGIC.SPATIAL.03 · Larger stacks

Four discs with centers lined up are stacked, bottom to top: radius 6, 2, 5, 3 (cm). How many can be seen from above?

Answer: A. 3

Reasoning: Visible: 6 (nothing larger above), 5 (only 3 above), 3 (top). The 2 cm disc is under the 5 cm disc.

- ✗ 4: the 2 cm disc is covered by the 5 cm disc above it
- ✗ 2: the 5 cm disc shows as a ring outside the 3 cm disc
- ✗ 1: larger discs below show around smaller ones

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-F1-Q5 · question page 465

LOGIC.SPATIAL.03 · Larger stacks

If two sheets do not overlap at all, the picture from above cannot show which one is higher.

Answer: True

Reasoning: Nothing in the picture depends on their order.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S20-F2-Q1 · question page 466

LOGIC.SPATIAL.03 · How much does a picture tell?

Three sheets: A and B do not overlap each other, and sheet C overlaps both and is on top of both. How many different bottom-to-top orders are consistent with the picture from above?

Answer: 2

Reasoning: C must be on top. A and B can be in either order below it: A,

B, C or B, A, C.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F2-Q2 · question page 467

LOGIC.SPATIAL.03 · How much does a picture tell?

Discs of radius 1, 2, 3 and 4 cm are stacked with centers lined up. In which stacking order (bottom to top) can ALL four discs be seen from above?

Answer: B. 4, 3, 2, 1

Reasoning: Each disc must be larger than every disc above it: largest at the bottom.

- ✗ 1, 2, 3, 4: only the top 4 cm disc shows
- ✗ 4, 2, 3, 1: the 2 cm disc is under the 3 cm disc
- ✗ 3, 4, 2, 1: the 3 cm disc is under the 4 cm disc

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S20-F2-Q3 · question page 468

LOGIC.SPATIAL.03 · How much does a picture tell?

Discs of radius 1, 2, 3 and 4 cm are stacked in some order with centers lined up. Over all 24 stacking orders, in how many is EXACTLY one disc visible from above?

Answer: 6

Reasoning: Exactly one disc shows when the 4 cm disc is on top. The other three can be in any order below: 3 x 2 x 1 = 6.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S20-F2-Q4 · question page 469

LOGIC.SPATIAL.03 · How much does a picture tell?

From a picture of a stack of sheets, the TOP sheet can always be identified if every sheet overlaps at least one other sheet.

Answer: False

Reasoning: Two separate piles (A under B, and C under D, with no overlap between the piles) have two sheets with complete outlines, B and D, and the picture cannot tell which of them is the highest.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S20-F2-Q5 · question page 470

LOGIC.SPATIAL.03 · How much does a picture tell?

Cards on a grid: P covers rows 1-3, columns 1-3; Q covers rows 2-4, columns 2-4. P was placed first. Which cell shows card P?

Answer: A. row 1, column 2

Reasoning: Q covers rows 2-4 and columns 2-4, so P shows in row 1 and in column 1.

- ✗ row 3, column 3: covered by Q
- ✗ row 2, column 2: covered by Q
- ✗ row 4, column 4: that cell is only Q

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-M-Q1 · question page 472

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Premises: "If the reservoir is full, the dam gates open." "The reservoir is full." What conclusion follows?

Answer: D. The dam gates open.

Reasoning: Modus ponens: if P then Q; P; therefore Q.

- ✗ The reservoir is not full.: contradicts a premise
- ✗ The dam gates do not open.: contradicts the conditional
- ✗ Nothing follows.: modus ponens gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the form was checked by truth table.

S21-M-Q2 · question page 473

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Premises: "If a number is divisible by 9, it is divisible by 3." "The number 52 is not divisible by 3." What follows?

Answer: C. 52 is not divisible by 9.

Reasoning: Modus tollens: if P then Q; not Q; therefore not P.

- ✗ 52 is divisible by 9.: contradicts the premises
- ✗ 52 is divisible by 3.: contradicts a premise
- ✗ Nothing follows.: modus tollens gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S21-M-Q3 · question page 474

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Arrange the steps of the argument so that each line follows from earlier ones and the conclusion comes last.

Answer: If the trail is icy, the park closes. → The trail is icy. → So the park closes. → If the park closes, the race is cancelled. → So the race is cancelled.

Also accepted: If the trail is icy, the park closes. → If the park closes, the race is cancelled. → The trail is icy. → So the park closes. → So the race is cancelled. | If the trail is icy, the park closes. → The trail is icy. → If the park closes, the race is cancelled. → So the park closes. → So the race is cancelled. | If the park closes, the race is cancelled. → If the trail is icy, the park closes. → The trail is icy. → So the park closes. → So the race is cancelled. | If the park closes, the race is cancelled. → The trail is icy. → If the trail is icy, the park closes. → So the park closes. → So the race is cancelled. | The trail is icy. → If the trail is icy, the park closes. → If the park closes, the race is cancelled. → So the park closes. → So the race is cancelled. | The trail is icy. → If the trail is icy, the park closes. → So the park closes. → If the park closes, the race is cancelled. → So the race is cancelled. | The trail is icy. → If the park closes, the race is cancelled. → If the trail is icy, the park closes. → So the park closes. → So the race is cancelled.

Reasoning: D needs A and C before it; E needs B and D before it and comes last. The premises A, B and C may be stated in any order that respects this, so all 8 such orders are accepted.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 8 valid.

S21-M-Q4 · question page 475

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

Premises: "The key is in the drawer or in the coat." "The key is not in the drawer." What follows?

Answer: B. The key is in the coat.

Reasoning: Disjunctive syllogism: P or Q; not P; therefore Q.

- ✗ The key is in the drawer.: contradicts a premise
- ✗ The key is in neither place.: contradicts the first premise
- ✗ Nothing follows.: disjunctive syllogism gives a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-M-Q5 · question page 476

LOGIC.INFERENCE.01 · Draw valid conclusions with modus ponens, modus tollens, chains and disjunctive syllogism

This argument is valid: "If P then Q. If Q then R. So if P then R."

Answer: True

Reasoning: If P is true, the first premise gives Q and the second gives R (a chain, or hypothetical syllogism).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-B1-Q1 · question page 477

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is false when P is false and Q is true.

Answer: False

Reasoning: With P false the conditional makes no promise, so it is true. It is false only when P is true and Q is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B1-Q2 · question page 478

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is false when P is false.

Answer: False

Reasoning: It is true whenever P is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B1-Q3 · question page 479

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is true and P is true. What is Q?

Answer: A. true

Reasoning: The only row with P true and the conditional true has Q true.

- ✗ false: then the conditional would be false
- ✗ either: Q is forced
- ✗ neither: Q has a value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B1-Q4 · question page 480

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is true and Q is false. What is P?

Answer: C. false

Reasoning: The only row with Q false and the conditional true has P false.

- ✗ true: then the conditional would be false
- ✗ either: P is forced
- ✗ neither: P has a value

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B1-Q5 · question page 481

LOGIC.CONDITIONAL.01 · When is a conditional true?

"If P then Q" is true and Q is true. P must be true.

Answer: False

Reasoning: P can be false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B2-Q1 · question page 482

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"P or Q" is true and Q is false. Then P is true.

Answer: True

Reasoning: At least one part is true, and it is not Q, so P is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B2-Q2 · question page 483

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"P or Q" is true and P is true. Then Q is false.

Answer: False

Reasoning: Inclusive or allows both.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-B2-Q3 · question page 484

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"The meeting is on Monday or Tuesday." It is not on Monday. When is it?

Answer: B. Tuesday

Reasoning: Eliminate Monday.

- ✗ Monday: ruled out
- ✗ neither day: the premise says one of them
- ✗ it cannot be told: only Tuesday is left

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B2-Q4 · question page 485

LOGIC.CONNECTIVE.02 · Using "or" and "not"

Which is the negation of "the light is on"?

Answer: D. the light is not on

Reasoning: Put "not" on the claim.

✗ the light is dim: not a negation
✗ the lamp is broken: unrelated
✗ the light is bright: not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S21-B2-Q5 · question page 486

LOGIC.CONNECTIVE.02 · Using "or" and "not"

"P or Q" is true and both P and Q could be true.

Answer: True

Reasoning: Inclusive or.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S21-F1-Q1 · question page 487

LOGIC.INFERENCE.01 · Longer chains of valid steps

All three premises are about the same day. Premises: "If today is a holiday, the bank is closed today." "If the bank is closed today, Mara cannot make a deposit there today." "Mara made a deposit there today." What follows?

Answer: B. Today is not a holiday.

Reasoning: Mara made a deposit there today, so the bank was not closed today (modus tollens); so today is not a holiday (modus tollens again).

✗ Today is a holiday.: contradicts the premises
✗ The bank is closed today.: contradicts the premises
✗ Nothing follows.: two steps of modus tollens give a conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-F1-Q2 · question page 488

LOGIC.INFERENCE.01 · Longer chains of valid steps

Valid: "P or Q. If P then R. If Q then R. So R."

Answer: True

Reasoning: Either way R follows (proof by cases).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F1-Q3 · question page 489

LOGIC.INFERENCE.01 · Longer chains of valid steps

Premises: "Ana or Ben will host." "If Ben hosts, the party is on Friday." "The party is not on Friday." Who hosts?

Answer: C. Ana

Reasoning: Not Friday, so Ben does not host (modus tollens); Ana or Ben hosts, so Ana does (disjunctive syllogism).

- ✗ Ben: then the party would be on Friday
- ✗ both: Ben does not host
- ✗ it cannot be told: the premises decide it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-F1-Q4 · question page 490

LOGIC.INFERENCE.01 · Longer chains of valid steps

Arrange into a valid argument ending with the conclusion "The fountain is off."

Answer: If the fountain is on, the pump runs. → The pump does not run. → So the fountain is off.

Also accepted: The pump does not run. → If the fountain is on, the pump runs. → So the fountain is off.

Reasoning: Modus tollens: the rule and the fact (in either order), then the conclusion.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 2 valid.

S21-F1-Q5 · question page 491

LOGIC.INFERENCE.01 · Longer chains of valid steps

Valid: "If P then Q. Not Q. So not P."

Answer: True

Reasoning: Modus tollens.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F2-Q1 · question page 492

LOGIC.INFERENCE.01 · Valid forms with several variables

Valid: "If P then (Q and R). Not R. So not P."

Answer: True

Reasoning: If P were true, R would be true; R is false, so P is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F2-Q2 · question page 493

LOGIC.INFERENCE.01 · Valid forms with several variables

Valid: "If P then (Q or R). Not Q. So R."

Answer: False

Reasoning: P may be false: P false, Q false, R false makes both premises true and the conclusion false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S21-F2-Q3 · question page 494

LOGIC.INFERENCE.01 · Valid forms with several variables

Premises: "If A then B." "If C then D." "A or C." Which conclusion follows?

Answer: A. B or D

Reasoning: If A, then B; if C, then D; one of A, C holds, so one of B, D holds (constructive dilemma).

- ✗ B and D: only one of A and C need be true
- ✗ A and C: only one is given
- ✗ not B: contradicts a possible case

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S21-F2-Q4 · question page 495

LOGIC.INFERENCE.01 · Valid forms with several variables

An argument has premises "P or Q" and "if P then R" and conclusion "Q or R". In how many of the 8 rows of its truth table are both premises true?

Answer: 4

Reasoning: Count the rows where P or Q is true and P -> R is true: TTT, FTT, FTF, TFT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S21-F2-Q5 · question page 496

LOGIC.INFERENCE.01 · Valid forms with several variables

The argument "P or Q. If P then R. So Q or R." is valid.

Answer: True

Reasoning: In each row where both premises are true, Q or R is true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-M-Q1 · question page 498

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

"If a phone is waterproof, it survives a fall into a pool. Leo's phone survived a fall into a pool. So Leo's phone is waterproof." What is wrong with this argument?

Answer: D. It affirms the consequent: the phone may have survived for another reason.

Reasoning: From "if P then Q" and Q, P does not follow.

- ✗ Nothing: it is modus ponens.: modus ponens starts from the "if" part, not the "then" part
- ✗ It denies the antecedent.: no premise says "not waterproof"
- ✗ A premise is false.: the problem is the form, whatever the premises

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S22-M-Q2 · question page 499

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

"If n is a multiple of 4, then n is even. 6 is not a multiple of 4. So 6 is not even." What is wrong?

Answer: A. It denies the antecedent: not P does not give not Q.

Reasoning: The conclusion is false (6 is even) even though both premises are true, so the form is invalid.

- ✗ Nothing: it is modus tollens.: modus tollens starts from not Q
- ✗ It affirms the consequent.: no premise says "6 is even"
- ✗ The conditional is false.: it is true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S22-M-Q3 · question page 500

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

This form is valid: "If P then Q. Q. So P."

Answer: False

Reasoning: P false, Q true makes both premises true and the conclusion false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-M-Q4 · question page 501

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

Use whole numbers to show that "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." is invalid: give a value of n that makes both premises true and the conclusion false.

Answer: any values that satisfy: (isEven(n) and n % 6 != 0) and isInteger(n) and n >= 0 (example: n = 4)

Reasoning: Any even number that is not a multiple of 6 (2, 4, 8, 10, ...): the conditional is always true, n is even, and n is not a multiple of 6.

Tempting answers the rule rejects: n = 6; n = 3; n = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S22-M-Q5 · question page 502

LOGIC.INFERENCE.02 · Recognize affirming the consequent and denying the antecedent

Which argument is VALID?

Answer: D. "If it rains, the game is off. The game is not off. So it is not raining."

Reasoning: Only the modus tollens argument is valid.

- ✗ "If it rains, the game is off. The game is off. So it is raining.": affirms the consequent
- ✗ "If it rains, the game is off. It is not raining. So the game is not off.": denies the antecedent
- ✗ "If it rains, the game is off. So if the game is off, it rains.": concludes the converse

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B1-Q1 · question page 503

LOGIC.INFERENCE.01 · Valid forms first

"If P then Q. P. So Q." What is this form called?

Answer: C. modus ponens

Reasoning: Affirm the "if" part.

- ✗ modus tollens: that uses not Q
- ✗ affirming the consequent: that uses Q as a premise
- ✗ denying the antecedent: that uses not P

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B1-Q2 · question page 504

LOGIC.INFERENCE.01 · Valid forms first

"If P then Q. Not Q. So not P." What is this form called?

Answer: B. modus tollens

Reasoning: Deny the "then" part.

- ✗ modus ponens: that uses P
- ✗ denying the antecedent: that uses not P
- ✗ affirming the consequent: that uses Q

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B1-Q3 · question page 505

LOGIC.INFERENCE.01 · Valid forms first

Modus ponens is valid.

Answer: True

Reasoning: Truth table check.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-B1-Q4 · question page 506

LOGIC.INFERENCE.01 · Valid forms first

Modus tollens is valid.

Answer: True

Reasoning: Truth table check.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-B1-Q5 · question page 507

LOGIC.INFERENCE.01 · Valid forms first

In modus ponens, which part of the conditional is given as a fact?

Answer: A. the "if" part

Reasoning: P is given.

- ✗ the "then" part: that would affirm the consequent
- ✗ neither part: one part is given
- ✗ the negation of the "if" part: that would deny the antecedent

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S22-B2-Q1 · question page 508

LOGIC.QUANT.03 · A counterexample shows invalidity

An argument form is invalid if there is one case with all premises true and the conclusion false.

Answer: True

Reasoning: One such case is enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-B2-Q2 · question page 509

LOGIC.QUANT.03 · A counterexample shows invalidity

"If x > 5, then x > 2. x > 2. So x > 5." Which value of x shows it is invalid?

Answer: D. 3

Reasoning: x = 3: both premises true, conclusion false.

- ✗ 6: the conclusion is true
- ✗ 1: the premise x > 2 is false
- ✗ 10: the conclusion is true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S22-B2-Q3 · question page 510

LOGIC.QUANT.03 · A counterexample shows invalidity

A valid argument can have true premises and a false conclusion.

Answer: False

Reasoning: That is what validity rules out.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-B2-Q4 · question page 511

LOGIC.QUANT.03 · A counterexample shows invalidity

"If an animal is a robin, it has wings. A bat has wings. So a bat is a robin." Which fact shows the form is invalid?

Answer: C. A bat has wings but is not a robin.

Reasoning: True premises, false conclusion.

- ✗ Robins have wings.: that is a premise
- ✗ Bats fly at night.: irrelevant
- ✗ Some birds cannot fly.: irrelevant

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-B2-Q5 · question page 512

LOGIC.QUANT.03 · A counterexample shows invalidity

If one case with true premises has a true conclusion, the argument is valid.

Answer: False

Reasoning: Validity needs EVERY case with true premises to have a true conclusion; one good case is not enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-F1-Q1 · question page 513

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

"The alarm sounds only if there is smoke. There is smoke. So the alarm sounds." Is this valid?

Answer: B. No: "A only if S" means "if A then S"; from S, A does not follow.

Reasoning: This affirms the consequent of "if the alarm sounds, there is smoke".

- ✗ Yes: it is modus ponens.: "only if" puts the alarm in the "if" part
- ✗ Yes: smoke always sets off alarms.: the premises do not say that
- ✗ No: the conclusion is false.: validity is about form, not the truth of the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-F1-Q2 · question page 514

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

"Every square is a rectangle. This shape is not a square. So it is not a rectangle." This argument is valid.

Answer: False

Reasoning: Denying the antecedent: a non-square rectangle is a counterexample.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-F1-Q3 · question page 515

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

Show that "If $x > 10$, then $x > 3$. x is not greater than 10. So x is not greater than 3." is

invalid: give a number x that makes the premises true and the conclusion false.

Answer: any values that satisfy: not (x > 10) and x > 3 (example: x = 5)

Reasoning: Any x with $3 < x \leq 10$.

Tempting answers the rule rejects: x = 2; x = 11; x = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S22-F1-Q4 · question page 516

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

Which argument has the SAME form as "If it is a cat, it is a mammal. It is a mammal. So it is a cat."?

Answer: A. "If n is 2, n is even. n is even. So n is 2."

Reasoning: Both affirm the consequent.

- ✗ "If n is 2, n is even. n is 2. So n is even.": modus ponens
- ✗ "If n is 2, n is even. n is odd. So n is not 2.": modus tollens
- ✗ "n is 2 or n is 3. n is not 3. So n is 2.": disjunctive syllogism

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-F1-Q5 · question page 517

LOGIC.INFERENCE.02 · Spot invalid forms in disguise

Denying the antecedent and affirming the consequent are both invalid.

Answer: True

Reasoning: Both have counterexample rows.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-F2-Q1 · question page 518

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

"If a number is divisible by 4, it is even. 12 is even. So 12 is divisible by 4." The conclusion is true, so the argument is valid.

Answer: False

Reasoning: The conclusion happens to be true, but the form (affirming the consequent) is invalid; 6 shows the form can fail.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S22-F2-Q2 · question page 519

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

For whole numbers n from 1 to 30, the argument "If n is a multiple of 6, then n is even. n is even. So n is a multiple of 6." has true premises for every even n. For how many of these n is the conclusion false?

Answer: 10

Reasoning: Even numbers up to 30: 15. Multiples of 6: 5. So 10 even numbers are not multiples of 6.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S22-F2-Q3 · question page 520

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

Which argument is INVALID, even though every premise and the conclusion are true?

Answer: C. "If 9 is odd, then 9 is a whole number. 9 is a whole number. So 9 is odd."

Reasoning: The keyed argument affirms the consequent, so it is invalid even though every sentence in it is true.

- ✗ "If 9 is odd, then 9 is a whole number. 9 is odd. So 9 is a whole number.": valid (modus ponens)
- ✗ "9 is odd or 9 is even. 9 is not even. So 9 is odd.": valid (disjunctive syllogism)
- ✗ "If 9 is even, then 9 is a whole number. 9 is even. So 9 is a whole number.": valid (modus ponens); its second premise is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S22-F2-Q4 · question page 521

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

Valid: "If P then Q. If not P then Q. So Q."

Answer: True

Reasoning: Q follows whether P is true or false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S22-F2-Q5 · question page 522

LOGIC.INFERENCE.02 · Invalid or just unconvincing?

Valid: "If P then Q. If Q then P. Not P. So not Q."

Answer: True

Reasoning: The premises make P and Q equivalent; not P then gives not

Q.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S23-M-Q1 · question page 524

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is on the face OPPOSITE the face A?

Answer: B. E

Reasoning: In the cross-shaped net, the column A, C, E, F folds around the cube: A and E are two squares apart in the column, so they are opposite. (C is opposite F, and B is opposite D.)

- ✗ C: C shares an edge with A in the net, so they touch on the cube
- ✗ B: B folds up next to A, so they touch
- ✗ F: F folds over the top and touches A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-M-Q2 · question page 525

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

Each figure is made of six squares. Which figure can be folded along the lines between squares into a closed cube (with no overlapping squares)?

Answer: C. Figure C

Reasoning: Fold C: its row of four squares wraps around to make four sides, and the two single squares (one above and one below the row) close the top and the bottom. The others fold with two squares on the

same face.

- ✗ Figure A: five squares in a row: the 1st and 5th squares would land on the same face
- ✗ Figure B: a 2 by 3 block: the squares fold into a ring of four with two squares overlapping
- ✗ Figure D: four squares meet around one point, so two faces of the cube would overlap when folded

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S23-M-Q3 · question page 526

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face P, Q and R of this net, which letter is on the opposite face?

Answer: P: U; Q: S; R: T

Reasoning: In the row Q, R, S, T, squares two apart are opposite: Q-S and R-T. P and U, attached at opposite ends, are opposite each other.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S23-M-Q4 · question page 527

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which two letters can NEVER be seen at the same time when you look at the cube?

Answer: D. G and J

Reasoning: Two faces can never be seen together only if they are

opposite. In this staircase net, G is opposite J, H is opposite K, and I is opposite L.

- ✗ G and H: they share an edge in the net
- ✗ H and I: they share an edge
- ✗ H and J: H is opposite K, so H and J share an edge on the cube

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-M-Q5 · question page 528

LOGIC.SPATIAL.04 · Match a cube to its net: opposite and adjacent faces

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. On the cube made from this net, faces M and W share an edge.

Answer: True

Reasoning: Folding this net makes M opposite O, N opposite W, and V opposite X. M and W are not an opposite pair, so they share an edge.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S23-B1-Q1 · question page 529

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

A straight row of four squares, numbered 1 to 4, is part of a cube net. Which square is opposite square 1?

Answer: B. 3

Reasoning: Exactly one square between them.

- ✗ 2: shares an edge with 1

✗ 4: 4 is opposite 2

✗ 1: a face is not opposite itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S23-B1-Q2 · question page 530

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

Two squares that share an edge in a net always share an edge on the cube.

Answer: True

Reasoning: Folding keeps shared edges together.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S23-B1-Q3 · question page 531

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

How many pairs of opposite faces does a cube have?

Answer: 3

Reasoning: 6 faces in pairs: 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-B1-Q4 · question page 532

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

On a cube, how many faces share an edge with any one face?

Answer: A. 4

Reasoning: All faces except itself and the opposite face.

- ✗ 5: the opposite face does not touch it
- ✗ 3: four sides surround it
- ✗ 6: a face does not touch itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S23-B1-Q5 · question page 533

LOGIC.SPATIAL.04 · Strategy: skip one square to find the opposite face

Two faces that are opposite on the cube can be seen at the same time.

Answer: False

Reasoning: You can see at most three faces at once, and they all meet at one corner; opposite faces never meet.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S23-F1-Q1 · question page 534

LOGIC.SPATIAL.04 · Harder nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. Which letter is opposite H?

Answer: D. K

Reasoning: In the staircase, H is opposite K (H and K are two steps apart along the stair).

- ✗ G: shares an edge with H
- ✗ I: shares an edge with H

✗ J: J is opposite G

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-F1-Q2 · question page 535

LOGIC.SPATIAL.04 · Harder nets

Each of the four figures is made of six squares. How many of them can be folded into a closed cube?

Answer: 2

Reasoning: Folding each figure: in A and C two squares land on the same face of the cube, so they do not make a closed cube. B (a staircase) and D (a row of four with one square above and one below) fold into a cube. So 2.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-F1-Q3 · question page 536

LOGIC.SPATIAL.04 · Harder nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. For each face M, N and V of this net, which letter is on the opposite face?

Answer: M: O; N: W; V: X

Reasoning: M, N, O are a row of three, so M and O (one square apart) are opposite. V, W, X are a row of three, so V and X are opposite. That leaves N and W as the third pair.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 6 assignments enumerated; 1 satisfy every clue.

S23-F1-Q4 · question page 537

LOGIC.SPATIAL.04 · Harder nets

Every net of a cube has exactly 6 squares.

Answer: True

Reasoning: A cube has 6 faces.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S23-F1-Q5 · question page 538

LOGIC.SPATIAL.04 · Harder nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. In the cross-shaped net (A on top; B, C, D across; E and F below C), which two faces each share an edge with all of B, C and D?

Answer: C. A and E

Reasoning: The opposite pairs are A-E, B-D and C-F. A face touches B, C and D unless it is one of them or opposite one of them; that leaves A and E.

- ✗ A and F: F is opposite C, so F does not touch C
- ✗ E and F: F does not touch C
- ✗ B and D: B and D are opposite each other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-F2-Q1 · question page 539

LOGIC.SPATIAL.04 · Counting and designing nets

Count the different nets of a cube: shapes of six squares joined edge to edge that fold into a cube, where shapes that are turns or flips of each other count as the same. How many are there?

Answer: 11

Reasoning: There are 35 shapes of six squares; folding each one shows that exactly 11 make a cube.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-F2-Q2 · question page 540

LOGIC.SPATIAL.04 · Counting and designing nets

The faces of a cube are colored so that faces sharing an edge always have different colors. What is the least number of colors needed?

Answer: 3

Reasoning: Opposite faces never touch, so each opposite pair can share a color: 3 colors. Two colors are not enough: the three faces at a corner all touch each other.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S23-F2-Q3 · question page 541

LOGIC.SPATIAL.04 · Counting and designing nets

On a standard die, opposite faces add to 7. In a cross-shaped net with 1, 2, 3, 4 in the long column and 5, 6 on the two sides, the numbers can follow this rule.

Answer: False

Reasoning: In the long column, the 1st and 3rd squares are opposite (1 and 3) and so are the 2nd and 4th (2 and 4); neither pair adds to 7.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S23-F2-Q4 · question page 542

LOGIC.SPATIAL.04 · Counting and designing nets

The squares are folded along their shared edges to make a cube, with the letters on the outside. Faces that are opposite each other on the cube never touch. The numbers 1 to 6 are written on the cross-shaped net with A = 1, B = 2, C = 3, D = 5, E = 6 and F = 4. Which statement is true?

Answer: A. Every pair of opposite faces adds to 7.

Reasoning: Opposite pairs of this net are A-E (1, 6), B-D (2, 5) and C-F (3, 4): each adds to 7.

- ✗ Faces 1 and 2 are opposite.: they share an edge
- ✗ Faces 3 and 4 touch.: they are opposite
- ✗ Faces 2 and 6 are opposite.: 2 is opposite 5

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S23-F2-Q5 · question page 543

LOGIC.SPATIAL.04 · Counting and designing nets

A cube net can contain four squares in a 2 by 2 block.

Answer: False

Reasoning: Four squares around one point would fold onto only three faces at that corner, so two would overlap.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S24-M-Q1 · question page 545

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

"All mammals breathe air. Whales are mammals. So whales breathe air." How is this argument best described?

Answer: D. valid and sound

Reasoning: The form is valid and both premises are true, so the argument is sound.

- ✗ valid but not sound: both premises are true
- ✗ sound but not valid: a sound argument must be valid
- ✗ invalid: the conclusion is forced by the premises

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-M-Q2 · question page 546

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

"All birds can swim. A sparrow is a bird. So a sparrow can swim." How is this argument best described?

Answer: B. valid but not sound

Reasoning: The form is valid; the first premise is false, so the argument is not sound.

- ✗ valid and sound: "All birds can swim" is false
- ✗ invalid: IF the premises were true, the conclusion would have to be true
- ✗ sound but not valid: sound arguments are always valid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-M-Q3 · question page 547

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

An argument can be sound even if its conclusion is false.

Answer: False

Reasoning: Sound means valid with true premises, and then the conclusion must be true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-M-Q4 · question page 548

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

"If a number is divisible by 10, it is even. 14 is even. So 14 is divisible by 10." How is this described?

Answer: D. invalid, so it cannot be sound

Reasoning: Affirming the consequent is invalid; an invalid argument is never sound.

- ✗ valid and sound: the form affirms the consequent
- ✗ valid but not sound: the form is not valid
- ✗ sound but not valid: impossible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-M-Q5 · question page 549

LOGIC.INFERENCE.03 · Tell whether an argument is valid, and whether it is also sound

Which argument is SOUND?

Answer: B. "Every prime greater than 2 is odd. 13 is a prime greater than 2. So 13 is odd."

Reasoning: Only the first is valid with true premises.

- ✗ "Every prime is odd. 2 is prime. So 2 is odd.": the first premise is false (2 is even)
- ✗ "If 13 is odd, 13 is prime. 13 is prime. So 13 is odd.": invalid form (affirming the consequent)
- ✗ "Every odd number is prime. 9 is odd. So 9 is prime.": the first premise is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-B1-Q1 · question page 550

LOGIC.INFERENCE.01 · Is the form valid?

"If P then Q. P. So Q." is a valid form.

Answer: True

Reasoning: Modus ponens.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S24-B1-Q2 · question page 551

LOGIC.INFERENCE.01 · Is the form valid?

"If P then Q. Q. So P." is a valid form.

Answer: False

Reasoning: Affirming the consequent.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S24-B1-Q3 · question page 552

LOGIC.INFERENCE.01 · Is the form valid?

A valid argument guarantees what?

Answer: C. If the premises are true, the conclusion is true.

Reasoning: Definition.

- ✗ The premises are true.: validity says nothing about whether premises are true
- ✗ The conclusion is true.: only if the premises are true
- ✗ The argument is convincing.: persuasiveness is a different question

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-B1-Q4 · question page 553

LOGIC.INFERENCE.01 · Is the form valid?

"P or Q. P. So not Q." is a valid form.

Answer: False

Reasoning: With inclusive "or", P and Q can both be true, so the premises can be true while "not Q" is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B1-Q5 · question page 554

LOGIC.INFERENCE.01 · Is the form valid?

An argument with a true conclusion is always valid.

Answer: False

Reasoning: An invalid form can reach a true conclusion by luck.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q1 · question page 555

LOGIC.ARGUMENT.01 · Are the premises true?

"Every rectangle has four equal sides" is true.

Answer: False

Reasoning: A 2 by 5 rectangle does not have four equal sides.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q2 · question page 556

LOGIC.ARGUMENT.01 · Are the premises true?

"Every rectangle is a square" is true.

Answer: False

Reasoning: A 2 by 5 rectangle is not a square.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q3 · question page 557

LOGIC.ARGUMENT.01 · Are the premises true?

"Every multiple of 2 is a multiple of 4" is true.

Answer: False

Reasoning: 6 is a multiple of 2 but not of 4.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q4 · question page 558

LOGIC.ARGUMENT.01 · Are the premises true?

"Every prime number is odd" is true.

Answer: False

Reasoning: 2 is prime and even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-B2-Q5 · question page 559

LOGIC.ARGUMENT.01 · Are the premises true?

"Water freezes at 10 degrees Celsius at normal air pressure" is true.

Answer: False

Reasoning: At normal air pressure water freezes at 0 degrees Celsius; at 10 degrees it is liquid.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F1-Q1 · question page 560

LOGIC.INFERENCE.03 · Classify harder arguments

"Every whole number that is divisible by 4 is divisible by 8. 12 is divisible by 4. So 12 is divisible by 8." How is it described?

Answer: A. valid but not sound

Reasoning: Valid form; false premise; and here the conclusion is also false.

- ✗ valid and sound: the first premise is false: 4 is not divisible by 8
- ✗ invalid: the form is valid
- ✗ sound but not valid: impossible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F1-Q2 · question page 561

LOGIC.INFERENCE.03 · Classify harder arguments

A valid argument with a false conclusion can have all of its premises true.

Answer: False

Reasoning: In a valid argument, true premises force a true conclusion. So a false conclusion means some premise is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F1-Q3 · question page 562

LOGIC.INFERENCE.03 · Classify harder arguments

"No reptile has fur. A snake is a reptile. So a snake has no fur." How is it described?

Answer: D. valid and sound

Reasoning: Valid form, true premises.

- ✗ valid but not sound: both premises are true
- ✗ invalid: the conclusion follows

✗ sound but not valid: impossible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F1-Q4 · question page 563

LOGIC.INFERENCE.03 · Classify harder arguments

If an argument is invalid, its conclusion must be false.

Answer: False

Reasoning: Its conclusion may be true by accident; invalid only means it is not guaranteed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F1-Q5 · question page 564

LOGIC.INFERENCE.03 · Classify harder arguments

Which statement is TRUE?

Answer: A. Every sound argument is valid.

Reasoning: Sound = valid + true premises.

✗ Every valid argument is sound.: valid arguments can have false premises
✗ Every argument with true premises is sound.: it must also be valid
✗ Every argument with a true conclusion is sound.: it could be invalid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F2-Q1 · question page 565

LOGIC.INFERENCE.03 · Validity, soundness and truth together

Which combination is IMPOSSIBLE for an argument?

Answer: C. valid, all premises true, conclusion false

Reasoning: Validity rules out exactly this combination.

- ✗ valid, a false premise, conclusion true: possible: "All fish fly; robins are fish; so robins fly"
- ✗ invalid, all premises true, conclusion true: possible: affirming the consequent with true parts
- ✗ invalid, all premises true, conclusion false: possible: that is what invalid allows

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F2-Q2 · question page 566

LOGIC.INFERENCE.03 · Validity, soundness and truth together

An argument whose premises can never all be true at the same time (they contradict each other) is valid.

Answer: True

Reasoning: There is no case with all premises true and the conclusion false, so it is valid (but never sound).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S24-F2-Q3 · question page 567

LOGIC.INFERENCE.03 · Validity, soundness and truth together

Of the 8 combinations (valid or invalid; premises all true or not; conclusion true or false), how many are impossible for an argument?

Answer: 1

Reasoning: Only "valid, all premises true, conclusion false" is impossible; every other combination has an example (for instance "valid, false premise, false conclusion": "All cats fly; Tom is a cat; so Tom flies").

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author

S24-F2-Q4 · question page 568

LOGIC.INFERENCE.03 · Validity, soundness and truth together

"If x = 3, then x squared = 9. x squared = 9. So x = 3." For real x, this argument is:

Answer: B. invalid: x = -3 makes the premises true and the conclusion false

Reasoning: Affirming the consequent.

- ✗ valid: squaring is unique: squaring is not one-to-one
- ✗ sound: both premises can be true: soundness needs validity
- ✗ valid but not sound: the form is not valid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S24-F2-Q5 · question page 569

LOGIC.INFERENCE.03 · Validity, soundness and truth together

Adding a false premise to a sound argument keeps it valid.

Answer: True

Reasoning: Adding premises never turns a valid argument invalid; it only stops it from being sound.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S25-M-Q1 · question page 571

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. The row reads T, E, A, M. Move the leftmost tile to the far right end. Then swap the first two tiles. What is the row now, from left to right?

Answer: A → E → M → T

Reasoning: Moving T to the end gives E, A, M, T. Swapping the first two gives A, E, M, T.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S25-M-Q2 · question page 572

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn D, C, B, A into A, B, C, D?

Answer: 6

Reasoning: Each move fixes at most one pair that is in the wrong order. D, C, B, A has 6 such pairs (every pair), so at least 6 moves are needed, and 6 moves suffice (bubble the letters into place).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-M-Q3 · question page 573

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. Which single move turns R, O, P, E into O, P, E, R?

Answer: D. Move R from the left end to the right end.

Reasoning: Taking R off the left and putting it on the right leaves O, P, E, R.

- ✗ Swap R and O.: that gives O, R, P, E
- ✗ Swap the two middle tiles.: that gives R, P, O, E
- ✗ Move E from the right end to the left end.: that gives E, R, O, P

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S25-M-Q4 · question page 574

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. One move swaps two tiles that are next to each other. What is the least number of moves that turn C, A, B, E, D into A, B, C, D, E?

Answer: 3

Reasoning: Pairs out of order: (C, A), (C, B), (E, D): 3. Each swap changes that count by one, so 3 moves are needed and enough: swap C and A, swap C and B, swap E and D.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-M-Q5 · question page 575

LOGIC.SEQUENCE.01 · Carry out stated moves on a row of tokens and find moves that reach a target

Tiles in a row each show a different letter. Reversing the whole row twice puts every tile back where it started.

Answer: True

Reasoning: The second reversal undoes the first.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S25-B1-Q1 · question page 576

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. Row: K, I, T. Swap the first two tiles. What is the row?

Answer: I → K → T

Reasoning: K and I trade places.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S25-B1-Q2 · question page 577

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. Row: P, A, N. Move the leftmost tile to the far right. What is the row?

Answer: A → N → P

Reasoning: P goes to the end: A, N, P.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 1 valid.

S25-B1-Q3 · question page 578

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. How many pairs of tiles are in the wrong alphabetical order in C, B, A?

Answer: 3

Reasoning: (C, B), (C, A), (B, A).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-B1-Q4 · question page 579

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. Swapping two neighboring tiles twice returns the row to how it was.

Answer: True

Reasoning: The second swap undoes the first.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S25-B1-Q5 · question page 580

LOGIC.SEQUENCE.01 · Strategy: number the positions

Tiles in a row each show a different letter. One move swaps neighboring tiles. Least number of moves from B, A, C to A, B, C?

Answer: 1

Reasoning: Swap B and A.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q1 · question page 581

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn B, C, D, A into A, B, C, D?

Answer: 3

Reasoning: A must end up first, so B, C and D must each be moved behind it: move B, then C, then D. Fewer is impossible because each of B, C, D is in front of A and only a move can put it after A.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q2 · question page 582

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. One move takes ANY tile and puts it at the far right end. What is the least number of moves that turn A, C, D, B into A, B, C, D?

Answer: 2

Reasoning: Move C, then D: A, D, B, C, then A, B, C, D.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q3 · question page 583

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. Row: S, T, O, P. Move the tile in position 3 to the far left end, then swap the last two tiles. What is the row?

Answer: O → S → P → T

Reasoning: Moving O to the front gives O, S, T, P; swapping the last two gives O, S, P, T.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S25-F1-Q4 · question page 584

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. One move swaps neighboring tiles. What is the least number of moves that turn E, D, C, B, A into A, B, C, D, E?

Answer: 10

Reasoning: Every one of the 10 pairs is out of order.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F1-Q5 · question page 585

LOGIC.SEQUENCE.01 · Fewest moves under different rules

Tiles in a row each show a different letter. If one move swaps neighboring tiles, any arrangement of 4 tiles can be reached from any other.

Answer: True

Reasoning: Neighbor swaps can sort any row (as in bubble sort), and reversing the steps reaches any arrangement.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S25-F2-Q1 · question page 586

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move reverses any three tiles that are next to each other (for example A, B, C becomes C, B, A). Starting from A, B, C, D, E, can you reach B, A, C, D, E?

Answer: A. No: each move keeps every tile in an odd or an even position, and A would need to move from position 1 to position 2.

Reasoning: Reversing three tiles swaps the 1st and 3rd of them and keeps the middle one: every tile moves by 0 or 2 places, so it stays on odd or on even positions. A starts in position 1 (odd) and would need position 2.

- ✗ Yes: reverse A, B, C, then reverse B, A, ... in a few moves.: no sequence works
- ✗ Yes: any arrangement can be reached with enough moves.: only 12 of the 120 arrangements can be reached
- ✗ No: because each move changes three tiles at once.: that alone does not rule it out

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: all positions reachable by the move were searched. Automatic: the check function accepts only the keyed choice.

S25-F2-Q2 · question page 587

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move reverses any three neighboring tiles. Starting from A, B, C, D, E, how many different arrangements can be reached (counting the start)?

Answer: 12

Reasoning: Tiles in odd positions (A, C, E) can be put in any order among positions 1, 3, 5 (6 ways), and B, D in positions 2, 4 (2 ways): 6 x 2 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S25-F2-Q3 · question page 588

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move swaps neighboring tiles. The row C, A, D, B, E must become A, B, C, D, E. Give the least number of moves m.

Answer: any values that satisfy: m == 3 (example: m = 3)

Reasoning: Pairs out of order: (C, A), (C, B), (D, B): 3, so 3 moves are needed and enough.

Tempting answers the rule rejects: m = 2; m = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S25-F2-Q4 · question page 589

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move swaps ANY two tiles (not only neighbors). D, C, B, A can be turned into A, B, C, D in 2 moves.

Answer: True

Reasoning: Swap D and A, then swap C and B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S25-F2-Q5 · question page 590

LOGIC.SEQUENCE.01 · Moves that can never work

Tiles in a row each show a different letter. One move swaps ANY two tiles. What is the least number of moves that turn B, C, D, E, A into A, B, C, D, E?

Answer: 4

Reasoning: Each swap can put at most one tile into its final place unless it fixes two at once, which needs two tiles in each other's places; here the five tiles form one cycle, so 5 - 1 = 4 swaps are needed.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-M-Q1 · question page 592

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "All chess members have a bus pass."

Answer: True

Reasoning: The chess members are Ava and Dee, and both have a bus pass.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-M-Q2 · question page 593

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben

(16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member aged 16 has a bus pass."

Answer: False

Reasoning: The members aged 16 are Ben and Eli; neither has a bus pass.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-M-Q3 · question page 594

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is TRUE?

Answer: C. No art member has a bus pass.

Reasoning: The only art member, Eli, has no bus pass.

- ✗ All robotics members have a bus pass.: Ben does not
- ✗ Some chess member is 16.: the chess members are 15 and 17
- ✗ Every member is at least 16.: Ava and Cy are 15

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S26-M-Q4 · question page 595

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). How many members make "the member is 15 or has no bus pass" true?

Answer: 4

Reasoning: Aged 15: Ava, Cy. No bus pass: Ben, Eli. Together: 4 members (Dee is the only exception).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-M-Q5 · question page 596

LOGIC.QUANT.01 · Decide the truth of "all", "some" and "no" statements over a stated domain

The domain is the whole numbers 1 to 10. Which statement is FALSE?

Answer: B. Every number in the domain is less than 10.

Reasoning: 10 is in the domain and is not less than 10.

- ✗ Some number in the domain is prime.: 2, 3, 5 and 7 are prime
- ✗ No number in the domain is negative.: this is true: 1 to 10 are all positive
- ✗ Some number in the domain is a multiple of 7.: this is true: 7 is in the domain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S26-B1-Q1 · question page 597

LOGIC.QUANT.01 · Read "all", "some" and "no"

"Some students walk to school" is true when ...

Answer: B. at least one student walks to school

Reasoning: "Some" means at least one.

✗ every student walks to school: that is "all"
✗ exactly one student walks: "some" allows more
✗ most students walk: "some" does not mean most

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-B1-Q2 · question page 598

LOGIC.QUANT.01 · Read "all", "some" and "no"

"No student walks to school" is true when ...

Answer: C. not even one student walks

Reasoning: "No" means zero.

✗ some students do not walk: that is "not all"
✗ most students ride: still allows walkers
✗ exactly one student walks: one walker makes it false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-B1-Q3 · question page 599

LOGIC.QUANT.01 · Read "all", "some" and "no"

"All the lights are on" is true if exactly one light is off.

Answer: False

Reasoning: One light that is off is an exception, and one exception makes an "all" statement false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-B1-Q4 · question page 600

LOGIC.QUANT.01 · Read "all", "some" and "no"

"Some numbers in {2, 4, 6} are odd" is true.

Answer: False

Reasoning: None of them is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-B1-Q5 · question page 601

LOGIC.QUANT.01 · Read "all", "some" and "no"

"Every number in {3, 5, 7} is odd" is true.

Answer: True

Reasoning: All three are odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-B2-Q1 · question page 602

LOGIC.QUANT.01 · Stay inside the stated domain

The domain is {1, 2, 3}. "Every number in the domain is less than 4" is true.

Answer: True

Reasoning: Only 1, 2, 3 count.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-B2-Q2 · question page 603

LOGIC.QUANT.01 · Stay inside the stated domain

The domain is {10, 20, 30}. "Some number in the domain is odd" is true.

Answer: False

Reasoning: All three are even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-B2-Q3 · question page 604

LOGIC.QUANT.01 · Stay inside the stated domain

The domain is the days Monday to Friday. Which is true?

Answer: A. No day in the domain is a weekend day.

Reasoning: Saturday and Sunday are outside the domain.

- ✗ Some day in the domain is Sunday.: Sunday is not in the domain
- ✗ Every day in the domain starts with M.: only Monday
- ✗ Some day in the domain is Saturday.: not in the domain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-B2-Q4 · question page 605

LOGIC.QUANT.01 · Stay inside the stated domain

"All prime numbers are odd" is true when the domain is {3, 5, 7, 11}.

Answer: True

Reasoning: Within this domain, every prime is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-B2-Q5 · question page 606

LOGIC.QUANT.01 · Stay inside the stated domain

"All prime numbers are odd" is true when the domain is all whole numbers.

Answer: False

Reasoning: 2 is prime and even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-F1-Q1 · question page 607

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Every member without a bus pass is 16."

Answer: True

Reasoning: Ben and Eli have no bus pass; both are 16.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F1-Q2 · question page 608

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). "Some member with a bus pass is older than every robotics member."

Answer: True

Reasoning: Dee (17) has a bus pass and is older than Ben (16) and Cy (15).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F1-Q3 · question page 609

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is the whole numbers 1 to 20. How many numbers make "the number is even and not a multiple of 3" true?

Answer: 7

Reasoning: Even: 10 numbers. Even multiples of 3 (6, 12, 18): 3. So 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-F1-Q4 · question page 610

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). Which statement is FALSE?

Answer: D. No member aged 15 is in robotics.

Reasoning: Cy is 15 and in robotics.

- ✗ Every chess member is 15 or 17.: true: Ava 15, Dee 17
- ✗ Some member has no bus pass.: Ben, Eli
- ✗ All members aged 17 play chess.: Dee is the only 17-year-old

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the

check function accepts only the keyed choice.

S26-F1-Q5 · question page 611

LOGIC.QUANT.01 · Quantifiers with two conditions

The domain is empty (it has no members). "Every member of the domain is green" is true.

Answer: True

Reasoning: There is no member to break the claim, so an "every" statement about an empty domain is true ("vacuously" true).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S26-F2-Q1 · question page 612

LOGIC.QUANT.01 · Nested quantifiers

The domain is the whole numbers 1 to 5. "For every number x, there is a number y with y > x" is true.

Answer: False

Reasoning: For x = 5 there is no larger number in the domain.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F2-Q2 · question page 613

LOGIC.QUANT.01 · Nested quantifiers

The domain is the whole numbers 1 to 5. "There is a number y such that for every number x, y ≥ x" is true.

Answer: True

Reasoning: y = 5 works for every x.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S26-F2-Q3 · question page 614

LOGIC.QUANT.01 · Nested quantifiers

The domain is all whole numbers. Which statement is TRUE?

Answer: C. For every x, there is a y with y > x.

Reasoning: Given any x, y = x + 1 works. The order of "for every" and "there is" matters.

- ✗ There is a y such that for every x, y > x.: no whole number is larger than every whole number (including itself)
- ✗ For every x and every y, y > x.: false for x = 3, y = 2
- ✗ There is an x such that for every y, y > x.: x would have to be less than itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S26-F2-Q4 · question page 615

LOGIC.QUANT.01 · Nested quantifiers

The domain is exactly these five club members: Ava (age 15, chess, has a bus pass), Ben (16, robotics, no bus pass), Cy (15, robotics, bus pass), Dee (17, chess, bus pass), Eli (16, art, no bus pass). For how many members x is the statement "some other member is the same age as x" true?

Answer: 4

Reasoning: Ava and Cy share 15; Ben and Eli share 16; Dee (17) has no partner.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S26-F2-Q5 · question page 616

LOGIC.QUANT.01 · Nested quantifiers

The domain is the whole numbers 1 to 9. Give a number x that makes "for every y in the domain, x + y > 5" true.

Answer: any values that satisfy: (x >= 5 and x <= 9 and isInteger(x)) and isInteger(x) and x >= 0 (example: x = 6)

Reasoning: The smallest y is 1, so x + 1 > 5, x > 4: any x from 5 to 9.

Tempting answers the rule rejects: x = 4; x = 10; x = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S27-M-Q1 · question page 618

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

What is the negation of "All the tickets were sold"?

Answer: A. At least one ticket was not sold.

Reasoning: The negation of "all are" is "some (at least one) is not".

- ✗ No ticket was sold.: too strong: one unsold ticket is enough to make the original false
- ✗ All the tickets were unsold.: same as "no ticket was sold"
- ✗ Some tickets were sold.: compatible with the original

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S27-M-Q2 · question page 619

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

What is the negation of "Some student brought a calculator"?

Answer: D. No student brought a calculator.

Reasoning: The negation of "at least one is" is "none is".

- ✗ Some student did not bring a calculator.: can be true together with the original
- ✗ Every student brought a calculator.: that makes the original true
- ✗ Exactly one student brought a calculator.: that makes the original true

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-M-Q3 · question page 620

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

What is the negation of "No bridge in the town is closed"?

Answer: B. Some bridge in the town is closed.

Reasoning: The negation of "none is" is "at least one is".

- ✗ Every bridge in the town is closed.: too strong
- ✗ Some bridge in the town is open.: compatible with the original
- ✗ No bridge in the town is open.: not the negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-M-Q4 · question page 621

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

The domain is the whole numbers. The statement "Every whole number greater than 1 has an even square" is false. Give a number n that makes its negation true: a whole number greater than 1 whose square is odd.

Answer: any values that satisfy: (n > 1 and isOdd(n*n) and isInteger(n)) and isInteger(n) and n >= 0 (example: n = 3)

Reasoning: The negation is "Some whole number greater than 1 has an odd square": any odd n > 1.

Tempting answers the rule rejects: n = 2; n = 1; n = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S27-M-Q5 · question page 622

LOGIC.QUANT.02 · Negate statements with "all", "some", "no", and nested quantifiers

The negation of "Every square is a rectangle" is "No square is a rectangle".

Answer: False

Reasoning: The negation is "Some square is not a rectangle".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q1 · question page 623

LOGIC.QUANT.01 · The meaning of each quantifier

"Some cards are red" is false if exactly one card is red.

Answer: False

Reasoning: "Some" means at least one, so one red card makes the statement true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q2 · question page 624

LOGIC.QUANT.01 · The meaning of each quantifier

"All cards are red" is false if one card is blue.

Answer: True

Reasoning: One exception is enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q3 · question page 625

LOGIC.QUANT.01 · The meaning of each quantifier

"No card is red" is true if one card is red.

Answer: False

Reasoning: "No" means zero.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B1-Q4 · question page 626

LOGIC.QUANT.01 · The meaning of each quantifier

"Not all cards are red" means ...

Answer: D. at least one card is not red

Reasoning: One non-red card is enough.

✗ no card is red: too strong

✗ exactly one card is red: unrelated
✗ every card is blue: too strong

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-B1-Q5 · question page 627

LOGIC.QUANT.01 · The meaning of each quantifier

"Some cards are red" and "some cards are not red" can both be true.

Answer: True

Reasoning: A mix of red and blue.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B2-Q1 · question page 628

LOGIC.CONNECTIVE.01 · Negating simple statements

What is the negation of "the bus is late"?

Answer: A. the bus is not late

Reasoning: Negate the whole claim.

✗ the bus is early: it could be on time
✗ the train is late: unrelated
✗ the bus is very late: not a negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-B2-Q2 · question page 629

LOGIC.CONNECTIVE.01 · Negating simple statements

The negation of "x ≥ 4" is "x ≤ 4".

Answer: False

Reasoning: x = 4 makes both "x ≥ 4" and "x ≤ 4" true, so they are not negations. The negation is "x < 4".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B2-Q3 · question page 630

LOGIC.CONNECTIVE.01 · Negating simple statements

n is an integer. What is the negation of "n is even and positive"?

Answer: A. n is odd or n is not positive

Reasoning: Not (A and B) = not A or not B. For an integer, "not even" means odd.

- ✗ n is odd and negative: too strong: n = 3 makes the original false but this false too
- ✗ n is even or positive: not a negation: n = 4 makes both true
- ✗ n is zero: too narrow: n = 3 also makes the original false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-B2-Q4 · question page 631

LOGIC.CONNECTIVE.01 · Negating simple statements

"Not (the door is locked)" means the door is not locked.

Answer: True

Reasoning: Plain negation.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-B2-Q5 · question page 632

LOGIC.CONNECTIVE.01 · Negating simple statements

What is the negation of "it rained on Monday or Tuesday"?

Answer: B. it rained on neither Monday nor Tuesday

Reasoning: Not (A or B) = not A and not B.

- ✗ it did not rain on Monday: it could have rained on Tuesday
- ✗ it rained on both days: not a negation
- ✗ it rained on Wednesday: unrelated

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F1-Q1 · question page 633

LOGIC.QUANT.02 · Negate quantified statements with conditions

What is the negation of "Every student who studied passed"?

Answer: B. Some student who studied did not pass.

Reasoning: Negation of "every A is B": some A is not B.

- ✗ Every student who did not study failed.: unrelated to the original claim
- ✗ No student who studied passed.: too strong
- ✗ Some student who passed did not study.: swaps the condition

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F1-Q2 · question page 634

LOGIC.QUANT.02 · Negate quantified statements with conditions

The domain is the real numbers. What is the negation of "For every x, x squared ≥ 0"?

Answer: C. There is an x with x squared < 0.

Reasoning: Negation: "there is an x" with the opposite inequality.

- ✗ For every x, x squared < 0.: too strong
- ✗ There is an x with x squared ≥ 0.: compatible with the original
- ✗ For every x, x squared ≤ 0.: not the negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F1-Q3 · question page 635

LOGIC.QUANT.02 · Negate quantified statements with conditions

The negation of "Some prime number is even" is "Every prime number is odd" (for whole numbers greater than 1).

Answer: True

Reasoning: "No prime is even" means every prime is not even; for whole numbers greater than 1, not even means odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-F1-Q4 · question page 636

LOGIC.QUANT.02 · Negate quantified statements with conditions

The statement "Every whole number from 1 to 30 that ends in 9 is prime" is false. Give a number that shows it (a witness for its negation).

Answer: any values that satisfy: (n % 10 == 9 and n >= 1 and n <= 30 and not isPrime(n)) and isInteger(n) and n >= 1 (example: n = 9)

Reasoning: The numbers ending in 9 are 9, 19 and 29; 19 and 29 are prime, but 9 = 3 x 3 is not. So n = 9.

Tempting answers the rule rejects: n = 19; n = 29; n = 39

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S27-F1-Q5 · question page 637

LOGIC.QUANT.02 · Negate quantified statements with conditions

The negation of "No dog is a cat" is "Every dog is a cat".

Answer: False

Reasoning: The negation is "Some dog is a cat".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-F2-Q1 · question page 638

LOGIC.QUANT.02 · Nested quantifiers and their negations

What is the negation of "For every student, there is a club the student likes"?

Answer: A. There is a student who likes no club.

Reasoning: Negate step by step: "there is a student such that NOT (there is a club the student likes)" = "there is a student who likes no club".

- ✗ For every student, there is a club the student does not like.: only flips the inner part
- ✗ There is a club that no student likes.: swaps the roles
- ✗ No student likes any club.: too strong

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F2-Q2 · question page 639

LOGIC.QUANT.02 · Nested quantifiers and their negations

The domain is the whole numbers. What is the negation of "There is a number y such that for every x, y > x"?

Answer: D. For every y, there is an x with y ≤ x.

Reasoning: Flip each quantifier and negate the inside: for every y, there is an x such that not (y > x).

- ✗ For every y and every x, y ≤ x.: too strong
- ✗ There is a y with y ≤ x for some x.: not the negation
- ✗ There is no number y.: not the negation

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S27-F2-Q3 · question page 640

LOGIC.QUANT.02 · Nested quantifiers and their negations

The negation of "For every x there is a y with x + y = 0" is "There is an x such that for every y, x + y ≠ 0".

Answer: True

Reasoning: Flip both quantifiers and negate the equation.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S27-F2-Q4 · question page 641

LOGIC.QUANT.02 · Nested quantifiers and their negations

The domain is the whole numbers 1 to 10. The statement "For every x there is a y with y = 2x" is false in this domain. Give an x that shows it.

Answer: any values that satisfy: (x >= 6 and x <= 10 and isInteger(x)) and isInteger(x) and x >= 0 (example: x = 6)

Reasoning: For x from 6 to 10, 2x is at least 12, outside the domain. Any of 6 to 10 works.

Tempting answers the rule rejects: x = 5; x = 1; x = 11

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S27-F2-Q5 · question page 642

LOGIC.QUANT.02 · Nested quantifiers and their negations

For any statement about a domain, either the statement or its negation is true.

Answer: True

Reasoning: A statement and its negation always have opposite truth values.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S28-M-Q1 · question page 644

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from A to F?

Answer: 3

Reasoning: A to B (or C), then to D, then to F: 3 roads. No route uses

fewer, because A is not joined to D or F, and no town next to A is joined to F.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-M-Q2 · question page 645

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many different routes go from A to F without visiting any town twice?

Answer: 8

Reasoning: Every route goes through D (the only way to E and F). From A to D: A-B-D, A-C-D, A-B-C-D, A-C-B-D (4 ways). From D to F: D-F or D-E-F (2 ways). 4 x 2 = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-M-Q3 · question page 646

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. Which town, if it were closed (with all its roads), would cut the network into two parts that cannot reach each other?

Answer: D. D

Reasoning: Every route from A, B or C to E or F passes through D.

- ✗ B: A can still reach D through C
- ✗ C: A can still reach D through B
- ✗ E: F is still joined to D

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-M-Q4 · question page 647

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Walkers 1, 2 and 3 each follow their own line from the left to a gate on the right. Where lines cross, there is no turning: each walker keeps going straight along their own line. Which gate does walker 2 reach?

Answer: A. Z

Reasoning: Follow line 2: it goes right, climbs to the top, crosses down to the bottom right and ends at gate Z.

- ✗ X: that is where another walker's line ends
- ✗ Y: that is where another walker's line ends
- ✗ none of the gates: every line ends at a gate

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-M-Q5 · question page 648

LOGIC.GRAPH.01 · Trace routes in a network and decide which routes exist

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once (towns may be visited more than once).

Answer: True

Reasoning: Count the roads at each town: A 2, B 3, C 3, D 4, E 2, F 2. Exactly two towns (B and C) have an odd number, so a walk using every road once exists: it must start at B and end at C (or the reverse).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S28-B1-Q1 · question page 649

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many roads meet at town D?

Answer: 4

Reasoning: D-B, D-C, D-E, D-F.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-B1-Q2 · question page 650

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. Which towns can be reached from A using exactly one road?

Answer: C. B and C

Reasoning: A is joined only to B and C.

- ✗ B, C and D: D needs two roads
- ✗ only B: C is also joined to A
- ✗ D: not joined to A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-B1-Q3 · question page 651

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. There is a road directly from A to D.

Answer: False

Reasoning: A is joined only to B and C.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S28-B1-Q4 · question page 652

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from E to F without visiting a town twice?

Answer: 2

Reasoning: E-F directly, or E-D-F.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-B1-Q5 · question page 653

LOGIC.GRAPH.01 · Strategy: mark the route as you trace it

In a network, a town with 3 roads must be the start or end of a walk that uses every road once.

Answer: True

Reasoning: At a town in the middle of the walk, roads are used in pairs (in and out); an odd number leaves one unpaired road, so the walk must start or end there.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S28-F1-Q1 · question page 654

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes go from P to U without visiting a town twice?

Answer: 6

Reasoning: Listing them in order: P-Q-R-U, P-Q-U, P-Q-T-U, P-S-T-U, P-S-T-Q-R-U, P-S-T-Q-U.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F1-Q2 · question page 655

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from P to U (no town twice) pass through T?

Answer: 4

Reasoning: P-Q-T-U, P-S-T-U, P-S-T-Q-R-U, P-S-T-Q-U.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F1-Q3 · question page 656

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. A walk can use every road exactly once.

Answer: True

Reasoning: Roads at each town: P 2, Q 4, R 2, S 2, T 3, U 3. Exactly two odd towns (T and U), so yes: start at T and end at U.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S28-F1-Q4 · question page 657

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. Which single road, if closed, makes T unreachable from P by any route?

Answer: B. none of these roads

Reasoning: T has three roads (to Q, S and U), and P reaches T through more than one of them, so no single road cuts T off.

- ✗ P-S: T can still be reached through Q
- ✗ Q-T: T can still be reached through S
- ✗ S-T: T can still be reached through Q

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-F1-Q5 · question page 658

LOGIC.GRAPH.01 · Routes with conditions

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads on a route from S to R?

Answer: 3

Reasoning: S-T-U-R or S-P-Q-R or S-T-Q-R: 3 roads.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F2-Q1 · question page 659

LOGIC.GRAPH.01 · Every road, every town

In a network of 6 towns, the numbers of roads at the towns are 1, 2, 2, 3, 3 and 3. Every road has two ends, each at a town. How many roads are there?

Answer: 7

Reasoning: Adding the numbers counts every road twice (once at each end): 1 + 2 + 2 + 3 + 3 + 3 = 14, so there are 14 / 2 = 7 roads.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F2-Q2 · question page 660

LOGIC.GRAPH.01 · Every road, every town

Some network has exactly three towns with an odd number of roads.

Answer: False

Reasoning: Every road adds 2 to the total count of road-ends, so the total is even; that needs an even number of odd towns.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S28-F2-Q3 · question page 661

LOGIC.GRAPH.01 · Every road, every town

Roads are the straight lines; towns are the circles. Roads meet only at towns. What is the least number of roads that must be ADDED so that a walk can use every road exactly once and return to its starting town?

Answer: B. 1

Reasoning: A closed walk needs every town to have an even number of roads. Only B and C are odd; adding one road B-C fixes both.

- ✗ 0: B and C have an odd number of roads, so a closed walk is impossible now
- ✗ 2: one new road between B and C makes every town even
- ✗ 3: far fewer are needed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S28-F2-Q4 · question page 662

LOGIC.GRAPH.01 · Every road, every town

Roads are the straight lines; towns are the circles. Roads meet only at towns. How many routes from A to F visit EVERY town exactly once?

Answer: 2

Reasoning: A route through all six towns must go A, then B and C (in some order), then D, then E, then F: A-B-C-D-E-F and A-C-B-D-E-F.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S28-F2-Q5 · question page 663

LOGIC.GRAPH.01 · Every road, every town

A network has towns with these numbers of roads: 3, 3, 2, 2 and k. For a walk that uses every road exactly once to be possible in a connected network, at most two towns may have an odd number of roads. Give a whole number k from 1 to 6 that keeps this possible.

Answer: any values that satisfy: (isEven(k) and k >= 1 and k <= 6) and

isInteger(k) and k >= 0 (example: k = 2)

Reasoning: There are already two odd towns (the two 3s), so k must be even: 2, 4 or 6.

Tempting answers the rule rejects: k = 3; k = 1; k = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 3 valid answers.

S29-M-Q1 · question page 665

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Give a witness for the true statement "Some whole number between 20 and 30 is prime."

Answer: any values that satisfy: (isPrime(n) and n > 20 and n < 30) and isInteger(n) and n >= 0 (example: n = 23)

Reasoning: The primes between 20 and 30 are 23 and 29.

Tempting answers the rule rejects: n = 21; n = 25; n = 31

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S29-M-Q2 · question page 666

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Give a counterexample to "Every whole number that is divisible by 3 is odd."

Answer: any values that satisfy: (n % 3 == 0 and isEven(n) and isInteger(n)) and isInteger(n) and n >= 0 (example: n = 6)

Reasoning: A counterexample is a multiple of 3 that is even: 6, 12, 18, ...

Tempting answers the rule rejects: n = 9; n = 4; n = 15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S29-M-Q3 · question page 667

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Which number is a counterexample to "Every number that ends in 1 is prime"?

Answer: A. 21

Reasoning: 21 ends in 1 and 21 = 3 x 7.

- ✗ 11: 11 is prime, so it agrees with the claim
- ✗ 31: 31 is prime
- ✗ 12: does not end in 1

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-M-Q4 · question page 668

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

Give integers a and b (integers are real numbers) that are a counterexample to "For all real numbers a and b, (a + b) squared = a squared + b squared."

Answer: any values that satisfy: (a != 0 and b != 0) and isInteger(a) and isInteger(b) (example: a = 1, b = 1)

Reasoning: (a + b) squared - (a squared + b squared) = 2ab, so the claim fails exactly when a and b are both nonzero. For example a = 1, b =

1: 4 is not 2.

Tempting answers the rule rejects: a = 0, b = 5; a = 3, b = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S29-M-Q5 · question page 669

LOGIC.QUANT.03 · Give a witness for a "some" statement and a counterexample to an "all" statement

One example where a statement is true proves "Every number has this property."

Answer: False

Reasoning: An example (a witness) proves only a "some" statement. An "every" statement needs an argument for all cases; one counterexample is enough to disprove it.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B1-Q1 · question page 670

LOGIC.QUANT.01 · Read "every" and "some" statements

What kind of example disproves "Every bird can fly"?

Answer: C. a bird that cannot fly

Reasoning: A counterexample has the "every" property but not the claimed one.

- ✗ a bird that can fly: agrees with the claim
- ✗ an animal that is not a bird: irrelevant
- ✗ a flying insect: irrelevant

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S29-B1-Q2 · question page 671

LOGIC.QUANT.01 · Read "every" and "some" statements

What kind of example proves "Some square numbers are even"?

Answer: D. an even square number, such as 4

Reasoning: One witness proves "some".

- ✗ an odd square number: does not help
- ✗ an even number that is not a square: not a square
- ✗ every square number: one example is enough

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S29-B1-Q3 · question page 672

LOGIC.QUANT.01 · Read "every" and "some" statements

"All cats are black" is shown true by one black cat.

Answer: False

Reasoning: One black cat shows that some cats are black. Showing that all cats are black needs every cat to be black.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B1-Q4 · question page 673

LOGIC.QUANT.01 · Read "every" and "some" statements

"All cats are black" is shown false by one white cat.

Answer: True

Reasoning: One counterexample.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B1-Q5 · question page 674

LOGIC.QUANT.01 · Read "every" and "some" statements

"No number is negative" is shown false by the number 1.

Answer: False

Reasoning: 1 is not negative, so it does not contradict the statement. A negative number such as -1 would.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B2-Q1 · question page 675

LOGIC.QUANT.03 · Test a number against a property

15 is prime.

Answer: False

Reasoning: 15 = 3 x 5.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S29-B2-Q2 · question page 676

LOGIC.QUANT.03 · Test a number against a property

27 is divisible by 3 and odd.

Answer: True

Reasoning: 27 = 3 x 9, odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S29-B2-Q3 · question page 677

LOGIC.QUANT.03 · Test a number against a property

What is 7 squared minus 7?

Answer: 42

Reasoning: 49 - 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S29-B2-Q4 · question page 678

LOGIC.QUANT.03 · Test a number against a property

2 is both prime and even.

Answer: True

Reasoning: 2 has exactly two factors and is even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-B2-Q5 · question page 679

LOGIC.QUANT.03 · Test a number against a property

Which number is a multiple of 4 but not of 8?

Answer: A. 12

Reasoning: 12 = 4 x 3.

- ✗ 16: a multiple of 8
- ✗ 8: a multiple of 8
- ✗ 10: not a multiple of 4

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-F1-Q1 · question page 680

LOGIC.QUANT.03 · Counterexamples in algebra

Give a real number x that is a counterexample to "For every real number x, x squared ≥ x."

Answer: any values that satisfy: x^2 < x (example: x = 0.5)

Reasoning: Any x strictly between 0 and 1, such as 0.5 (0.25 < 0.5).

Tempting answers the rule rejects: x = 2; x = -1; x = 1

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S29-F1-Q2 · question page 681

LOGIC.QUANT.03 · Counterexamples in algebra

Give positive whole numbers a and b that are a counterexample to "If a divides b x c, then a divides b or a divides c", using c = 2.

Answer: any values that satisfy: (divides(a, b * 2) and not divides(a, b) and not divides(a, 2) and a > 0 and b > 0 and isInteger(a) and isInteger(b)) and isInteger(a) and a >= 1 and isInteger(b) and b >= 1 (example: a = 4, b = 2)

Reasoning: a = 4, b = 2: 4 divides 2 x 2 = 4, but 4 divides neither 2 nor 2. (a = 6, b = 3 also works.)

Tempting answers the rule rejects: a = 3, b = 3; a = 2, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S29-F1-Q3 · question page 682

LOGIC.QUANT.03 · Counterexamples in algebra

Which value of n is a counterexample to "n squared + n + 1 is prime for every whole number n ≥ 1"?

Answer: B. 4

Reasoning: 4: 16 + 4 + 1 = 21 = 3 x 7.

- ✗ 1: 3 is prime
- ✗ 2: 7 is prime
- ✗ 3: 13 is prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-F1-Q4 · question page 683

LOGIC.QUANT.03 · Counterexamples in algebra

The statement "For every real x, if x > 1 then x squared > x" has no counterexample.

Answer: True

Reasoning: If x > 1, multiplying x > 1 by the positive x gives x squared > x.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S29-F1-Q5 · question page 684

LOGIC.QUANT.03 · Counterexamples in algebra

Give whole numbers a and b, each from 1 to 9, that show "a - b = b - a" can be false.

Answer: any values that satisfy: (a - b != b - a and a >= 1 and a <= 9 and b >= 1 and b <= 9) and isInteger(a) and a >= 1 and isInteger(b) and b >= 1 (example: a = 5, b = 2)

Reasoning: Any a different from b.

Tempting answers the rule rejects: a = 4, b = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 1 listed mistakes fail it.

S29-F2-Q1 · question page 685

LOGIC.QUANT.03 · Witnesses that are hard to find

Give a witness for "Some whole number from 1 to 100 leaves remainder 2 when divided by 3, remainder 3 when divided by 4, and remainder 4 when divided by 5."

Answer: any values that satisfy: (n % 3 == 2 and n % 4 == 3 and n % 5 == 4 and n >= 1 and n <= 100) and isInteger(n) and n >= 1 (example: n = 59)

Reasoning: Each remainder is one less than the divisor, so n + 1 is divisible by 3, 4 and 5, that is by 60. n + 1 = 60 gives n = 59, the only such number up to 100.

Tempting answers the rule rejects: n = 29; n = 119; n = 14

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the

stated domain: 1 valid answers.

S29-F2-Q2 · question page 686

LOGIC.QUANT.03 · Witnesses that are hard to find

Is there a whole number n from 1 to 100 with n squared ending in the digit 2?

Answer: D. No: squares end only in 0, 1, 4, 5, 6 or 9.

Reasoning: The last digit of n squared depends only on the last digit of n; checking 0-9 gives 0, 1, 4, 9, 6, 5, 6, 9, 4, 1.

- ✗ Yes: n = 12.: 144 ends in 4
- ✗ Yes: n = 22.: 484 ends in 4
- ✗ No: but only because 2 is even.: that is not the reason; 4 and 6 are even and do appear

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S29-F2-Q3 · question page 687

LOGIC.QUANT.03 · Witnesses that are hard to find

Give whole numbers a, b and c (each from 1 to 20) with a squared + b squared = c squared and a < b.

Answer: any values that satisfy: (a^2 + b^2 == c^2 and a < b and a >= 1 and a <= 20 and b >= 1 and b <= 20 and c >= 1 and c <= 20 and isInteger(a) and isInteger(b) and isInteger(c)) and isInteger(a) and a >= 1 and isInteger(b) and b >= 1 and isInteger(c) and c >= 1 (example: a = 3, b = 4, c = 5)

Reasoning: For example 3, 4, 5; 5, 12, 13; 6, 8, 10; 8, 15, 17; 9, 12, 15; 12, 16, 20.

Tempting answers the rule rejects: a = 2, b = 3, c = 4; a = 4, b = 3, c = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S29-F2-Q4 · question page 688

LOGIC.QUANT.03 · Witnesses that are hard to find

How many counterexamples to "Every odd number from 1 to 30 is prime" are there in that range?

Answer: 6

Reasoning: Odd numbers 1 to 29 that are not prime: 1, 9, 15, 21, 25, 27.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S29-F2-Q5 · question page 689

LOGIC.QUANT.03 · Witnesses that are hard to find

To disprove "There is a whole number n with n + 5 = n", one example is enough.

Answer: False

Reasoning: A "there is" statement is disproved by showing NO number works: here n + 5 = n would give 5 = 0, which is impossible for every n. A single example cannot do that.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-M-Q1 · question page 691

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Circle A is the set of students in soccer and circle B the set of students in band. Mark the region for students in soccer but NOT in band.

Answer: A

Reasoning: Inside A, outside B: the part of A that does not overlap B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-M-Q2 · question page 692

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Circle A is the set of even numbers and circle B the set of multiples of 3 (whole numbers 1 to 30 make up the outside box). Mark the region that contains 12.

Answer: AB

Reasoning: 12 is even and a multiple of 3: the overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-M-Q3 · question page 693

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Three circles: A (likes math), B (likes art), C (likes music). Mark every region for people who like EXACTLY two of the three subjects.

Answer: AB, AC, BC

Reasoning: The three regions where exactly two circles overlap, not the center.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-M-Q4 · question page 694

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

In a class of 30 students, 18 play soccer, 12 play in the band, and 5 do both. How many students do neither?

Answer: 5

Reasoning: Soccer or band: 18 + 12 - 5 = 25. Neither: 30 - 25 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-M-Q5 · question page 695

LOGIC.SETS.01 · Mark subset, intersection and complement relations on Venn diagrams

Every member of set A is also in set B (A is a subset of B). On a two-circle diagram, mark the region that must be EMPTY.

Answer: A

Reasoning: If every A is a B, nothing can be in A outside B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q1 · question page 696

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark the region for things in BOTH A and B.

Answer: AB

Reasoning: The overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q2 · question page 697

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark the region for things in NEITHER A nor B.

Answer: O

Reasoning: Outside both circles.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q3 · question page 698

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark every region for things in A (whether or not they are in B).

Answer: A, AB

Reasoning: A only and the overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B1-Q4 · question page 699

LOGIC.SETS.01 · Regions of a two-circle diagram

How many regions does a two-circle Venn diagram have, counting the outside?

Answer: 4

Reasoning: A only, B only, both, neither.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-B1-Q5 · question page 700

LOGIC.SETS.01 · Regions of a two-circle diagram

Mark every region for things in B but not in A.

Answer: B

Reasoning: B only.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S30-B2-Q1 · question page 701

LOGIC.CONNECTIVE.02 · Set words: and, or, not

The intersection of A and B contains things that are ...

Answer: C. in A and in B

Reasoning: Intersection = "and".

- ✗ in A or in B: that is the union
- ✗ in neither: outside both
- ✗ in A but not B: part of A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S30-B2-Q2 · question page 702

LOGIC.CONNECTIVE.02 · Set words: and, or, not

The union of A and B contains things that are ...

Answer: A. in A or in B (or both)

Reasoning: Union = inclusive "or".

- ✗ in A and in B: that is the intersection
- ✗ in exactly one: that excludes the overlap

✗ in neither: outside both

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S30-B2-Q3 · question page 703

LOGIC.CONNECTIVE.02 · Set words: and, or, not

The complement of A contains everything in the box that is not in A.

Answer: True

Reasoning: Complement = "not".

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-B2-Q4 · question page 704

LOGIC.CONNECTIVE.02 · Set words: and, or, not

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the union of A and B?

Answer: 5

Reasoning: {1, 2, 3, 4, 5}.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-B2-Q5 · question page 705

LOGIC.CONNECTIVE.02 · Set words: and, or, not

A = {1, 2, 3, 4}, B = {3, 4, 5}. How many numbers are in the intersection?

Answer: 2

Reasoning: {3, 4}.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F1-Q1 · question page 706

LOGIC.SETS.01 · Three-circle diagrams and counts

Three circles A, B, C. Mark every region for things in A or B but NOT in C.

Answer: A, B, AB

Reasoning: A only, B only and the A-B overlap outside C.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-F1-Q2 · question page 707

LOGIC.SETS.01 · Three-circle diagrams and counts

Three circles A, B, C. Mark every region for things in at least two of the sets.

Answer: AB, AC, BC, ABC

Reasoning: The three two-set overlaps and the center.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-F1-Q3 · question page 708

LOGIC.SETS.01 · Three-circle diagrams and counts

Of 40 students, 22 take French, 17 take Spanish and 9 take both. How many take exactly one of the two?

Answer: 21

Reasoning: French only 22 - 9 = 13; Spanish only 17 - 9 = 8; 13 + 8 = 21.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F1-Q4 · question page 709

LOGIC.SETS.01 · Three-circle diagrams and counts

Every member of A is in B, and no member of B is in C. On a three-circle diagram (A, B, C), mark every region that must be EMPTY.

Answer: A, AC, BC, ABC

Reasoning: "Every A is B" empties A outside B (A only and A-C only). "No B is C" empties every region in both B and C (B-C only and the center).

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S30-F1-Q5 · question page 710

LOGIC.SETS.01 · Three-circle diagrams and counts

If A is a subset of B, then the intersection of A and B is A.

Answer: True

Reasoning: Everything in A is also in B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-F2-Q1 · question page 711

LOGIC.SETS.01 · Inclusion and exclusion

In a group of 50 people, 25 like tea, 30 like coffee and 15 like both. How many like neither?

Answer: 10

Reasoning: Tea or coffee: 25 + 30 - 15 = 40. Neither: 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F2-Q2 · question page 712

LOGIC.SETS.01 · Inclusion and exclusion

Sets A, B, C have 10, 12 and 8 members. A and B share 4, A and C share 3, B and C share 2, and all three share 1. How many members are in the union?

Answer: 22

Reasoning: 10 + 12 + 8 - 4 - 3 - 2 + 1 = 22.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S30-F2-Q3 · question page 713

LOGIC.SETS.01 · Inclusion and exclusion

In a club of 20, s members swim and r members run; every member does at least one. Give possible numbers s and r if exactly 5 do both.

Answer: any values that satisfy: s + r - 5 == 20 and s >= 5 and r >= 5 and s <= 20 and r <= 20 and isInteger(s) and isInteger(r) (example: s = 12, r = 13)

Reasoning: Everyone does at least one, so s + r - 5 = 20, s + r = 25, with each at least 5.

Tempting answers the rule rejects: s = 10, r = 10; s = 20, r = 10

Grading: The learner's values are put into the rule; ANY values that make the rule true are

correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 16 valid answers.

S30-F2-Q4 · question page 714

LOGIC.SETS.01 · Inclusion and exclusion

Two sets can have 8 and 6 members and a union of 15 members.

Answer: False

Reasoning: The union has at most 8 + 6 = 14 members.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S30-F2-Q5 · question page 715

LOGIC.SETS.01 · Inclusion and exclusion

Sets A and B are inside a box of 30 things. A has 12 things, B has 9, and A and B share 4. How many things are in the box but in neither A nor B?

Answer: 13

Reasoning: 30 - (12 + 9 - 4) = 13.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-M-Q1 · question page 717

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region F have?

Answer: 6

Reasoning: F shares borders with B, D, E, G, H and I: 6 neighbors.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-M-Q2 · question page 718

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which region has the most neighbors?

Answer: B. F

Reasoning: Counting each region's neighbors: F has 6, more than any other.

- ✗ B: B has 5 neighbors
- ✗ D: D has 4 neighbors
- ✗ C: C has 3 neighbors

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S31-M-Q3 · question page 719

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions A and I are neighbors.

Answer: False

Reasoning: A borders only B, D and G.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S31-M-Q4 · question page 720

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Each region gets one color and neighbors must get different colors. What is the least number of colors needed for this map?

Answer: 3

Reasoning: R, S and T all border each other, so at least 3 colors are needed. Three are enough: P red, Q blue, R green, S blue, T red, U blue.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-M-Q5 · question page 721

LOGIC.GRAPH.02 · Turn a map into a neighbor graph and reason about neighbors and colorings

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Color each region red, blue or green so that neighbors differ. P is red and Q is blue. Give every region's color.

Answer: P: red; Q: blue; R: green; S: blue; T: red; U: blue

Also accepted: P: red; Q: blue; R: green; S: blue; T: red; U: green

Reasoning: R borders P and Q, so R is green. S borders P and R, so S is blue. T borders Q, R and S, so T is red. U borders only T, so U is blue or green; both are accepted.

Grading: The learner's complete assignment must equal the key or an accepted alternative.

Verified: Automatic: all 729 assignments enumerated; 2 satisfy every clue.

S31-B1-Q1 · question page 722

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region C have?

Answer: 3

Reasoning: C borders B, E and H.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-B1-Q2 · question page 723

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Regions B and D are neighbors.

Answer: True

Reasoning: They share a border.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S31-B1-Q3 · question page 724

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Two regions touch only at a single corner point. Are they neighbors under this rule?

Answer: C. No

Reasoning: A border must have length.

- ✗ Yes: a corner point is not a shared border
- ✗ Only if they are the same color: irrelevant
- ✗ It depends on their size: irrelevant

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S31-B1-Q4 · question page 725

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many neighbors does region U have?

Answer: 1

Reasoning: U borders only T.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-B1-Q5 · question page 726

LOGIC.GRAPH.02 · Strategy: draw the neighbor graph

If three regions all border each other, a coloring with different colors for neighbors needs at least 3 colors.

Answer: True

Reasoning: Each of the three must differ from the other two.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S31-F1-Q1 · question page 727

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the total of the neighbor counts of all nine regions?

Answer: 36

Reasoning: Each shared border is counted twice (once from each side), so the total is twice the number of shared borders.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F1-Q2 · question page 728

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many pairs of regions share a border?

Answer: 18

Reasoning: Half of the total of the neighbor counts.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F1-Q3 · question page 729

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. What is the least number of colors needed so that neighbors differ?

Answer: 3

Reasoning: B, D and F border each other, so at least 3 colors; a 3-coloring exists (checked over all colorings).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F1-Q4 · question page 730

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. In any coloring of this map with 3 colors where neighbors differ, regions A and F have the same color.

Answer: True

Reasoning: A borders B, D and G, and so does F. D borders both B and G. If B and G had different colors, B, D and G would use all three colors and A could not be colored. So B and G match, B, D, G use two colors, and A and F must both take the third color.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S31-F1-Q5 · question page 731

LOGIC.GRAPH.02 · Neighbor counts as clues

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. Which ONE region could be removed so that only 2 colors are needed for the rest?

Answer: D. R

Reasoning: Every group of three regions that all border each other contains R (P-Q-R, P-R-S, Q-R-T, R-S-T). Without R, the rest can alternate: P and T one color, Q, S and U the other.

- ✗ T: P, Q and R still border each other
- ✗ U: R, S and T still border each other
- ✗ S: P, Q and R still border each other

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S31-F2-Q1 · question page 732

LOGIC.GRAPH.02 · Colorings: count and prove

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can the map be colored with red, blue and green so that neighbors differ?

Answer: 12

Reasoning: Fix colors for P and Q (6 ways), then the rest follow as in the MAIN puzzle with 2 choices for U: 6 x 2 = 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F2-Q2 · question page 733

LOGIC.GRAPH.02 · Colorings: count and prove

Some map drawn on a page needs at least 4 colors so that neighbors differ.

Answer: True

Reasoning: For example, one region surrounded by three regions that all border each other: the four all border each other, so they need 4 different colors.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S31-F2-Q3 · question page 734

LOGIC.GRAPH.02 · Colorings: count and prove

A map has regions in a ring: region 1 borders 2, 2 borders 3, and so on, and the last region borders region 1 (and there are no other borders). Give a number of regions n, from 3 to 10, for which 2 colors are NOT enough.

Answer: any values that satisfy: isOdd(n) and n >= 3 and n <= 10 (example: n = 5)

Reasoning: Going around the ring the colors must alternate, which closes up only when n is even. Any odd n: 3, 5, 7 or 9.

Tempting answers the rule rejects: n = 4; n = 6; n = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 4 valid answers.

S31-F2-Q4 · question page 735

LOGIC.GRAPH.02 · Colorings: count and prove

Regions are neighbors when they share a border of some length; touching only at a corner point does not count. How many ways can this map be colored with red, blue and green so that neighbors differ, if A is red?

Answer: 2

Reasoning: Counted by checking all colorings with A red.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S31-F2-Q5 · question page 736

LOGIC.GRAPH.02 · Colorings: count and prove

If a map can be colored with 4 colors so that neighbors differ, it can also be colored with 3 colors.

Answer: False

Reasoning: Four regions that all border each other need 4 different colors, so 3 colors are not always enough.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-M-Q1 · question page 738

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

Premise: "No reptile is warm-blooded." Circle A is reptiles and circle B is warm-blooded animals. Mark the region this premise says is EMPTY.

Answer: AB

Reasoning: "No A is B" means the overlap is empty.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-M-Q2 · question page 739

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All violinists are musicians. All musicians practice. So all violinists practice." Is this valid?

Answer: C. Yes: every violinist is inside the musicians, who are all inside those who practice.

Reasoning: Nested circles: violinists inside musicians inside practicers.

- ✗ No: some violinists may not practice.: the premises rule that out
- ✗ Yes, but only if every musician is a violinist.: not needed
- ✗ No: the conclusion reverses a premise.: it does not

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-M-Q3 · question page 740

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All squares are rectangles. Some rectangles are not rhombuses. So some squares are not rhombuses." Is this valid?

Answer: A. No: the rectangles that are not rhombuses may all lie outside the squares.

Reasoning: In fact every square is a rhombus, so the premises are true and the conclusion false: invalid.

- ✗ Yes: squares are rectangles, so they share every property.: subsets do not share every property
- ✗ Yes: the conclusion is true.: validity is about form; and in fact every square IS a rhombus
- ✗ No: no rectangle is a square.: that is false: every square is a rectangle

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-M-Q4 · question page 741

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All A are B. Some C are A. So some C are B." This form is valid.

Answer: True

Reasoning: The C that is an A is inside B, so some C is B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-M-Q5 · question page 742

LOGIC.SETS.02 · Test categorical arguments with Venn diagrams, including the existence assumption

"All dragons breathe fire. So some dragons breathe fire." When does this follow?

Answer: D. Only if at least one dragon exists (the existence assumption).

Reasoning: From "all" to "some" needs at least one member.

- ✗ Always: "all" includes "some".: if there are no dragons, "all dragons breathe fire" is true but "some dragon breathes fire" is false
- ✗ Never: the premise is about all dragons.: it follows once a dragon exists
- ✗ Only if dragons are real animals.: it is about existence, not about being an animal

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-B1-Q1 · question page 743

LOGIC.SETS.01 · Mark statements on a Venn diagram

"All A are B." Mark the region that must be EMPTY.

Answer: A

Reasoning: Nothing in A outside B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q2 · question page 744

LOGIC.SETS.01 · Mark statements on a Venn diagram

"No A is B." Mark the region that must be EMPTY.

Answer: AB

Reasoning: The overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q3 · question page 745

LOGIC.SETS.01 · Mark statements on a Venn diagram

"Some A is B." Mark the region where something must be.

Answer: AB

Reasoning: At least one thing in the overlap.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q4 · question page 746

LOGIC.SETS.01 · Mark statements on a Venn diagram

"Some A is not B." Mark the region where something must be.

Answer: A

Reasoning: At least one thing in A outside B.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 4 regions tested against the condition.

S32-B1-Q5 · question page 747

LOGIC.SETS.01 · Mark statements on a Venn diagram

"All A are B" and "No A is B" can both be true if A has no members.

Answer: True

Reasoning: With A empty, both statements are true (vacuously).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-B2-Q1 · question page 748

LOGIC.INFERENCE.03 · Valid means no counterexample

An argument is valid if a situation exists with true premises and a false conclusion.

Answer: False

Reasoning: Such a situation shows it is invalid.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-B2-Q2 · question page 749

LOGIC.INFERENCE.03 · Valid means no counterexample

"All cats are mammals. All dogs are mammals. So all cats are dogs." Which situation shows it is invalid?

Answer: B. the real world: cats and dogs are both mammals but cats are not dogs

Reasoning: True premises, false conclusion.

- ✗ a world with no cats: then the conclusion is true
- ✗ a world where all mammals are dogs: then cats would be dogs
- ✗ a world with no mammals: then there are no cats

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S32-B2-Q3 · question page 750

LOGIC.INFERENCE.03 · Valid means no counterexample

"All A are B. All B are C. So all A are C." is valid.

Answer: True

Reasoning: Nested sets.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-B2-Q4 · question page 751

LOGIC.INFERENCE.03 · Valid means no counterexample

"Some A are B. Some B are C. So some A are C." is valid.

Answer: False

Reasoning: The A-and-B things and the B-and-C things may be different.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-B2-Q5 · question page 752

LOGIC.INFERENCE.03 · Valid means no counterexample

Which premise pair makes "So no A is C" follow?

Answer: C. "All A are B." and "No B is C."

Reasoning: A inside B, and B outside C, so A is outside C.

✗ "Some A are B." and "No B is C.": only some A are known to be outside C
✗ "All A are B." and "Some B are not C.": A may still overlap C

✗ "No A is B." and "No B is C.": A and C may overlap outside B

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-F1-Q1 · question page 753

LOGIC.SETS.02 · Harder categorical forms

"No A is B. All C are B. So no C is A." This form is valid.

Answer: True

Reasoning: C lies inside B and B is outside A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F1-Q2 · question page 754

LOGIC.SETS.02 · Harder categorical forms

"All A are B. No C is A. So no C is B." This form is valid.

Answer: False

Reasoning: C can overlap the part of B outside A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F1-Q3 · question page 755

LOGIC.SETS.02 · Harder categorical forms

"Some athletes are vegetarians. All vegetarians eat beans. So ..." Which conclusion follows?

Answer: D. some athletes eat beans

Reasoning: The athletes who are vegetarians eat beans.

✗ all athletes eat beans: only some are known to be vegetarians
✗ all bean eaters are athletes: reverses the relations
✗ no athlete eats meat: not supported

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-F1-Q4 · question page 756

LOGIC.SETS.02 · Harder categorical forms

Premises: "All A are B" and "No B is C". On a three-circle diagram (A, B, C), mark every region that the premises make EMPTY.

Answer: A, AC, BC, ABC

Reasoning: "All A are B" empties A outside B; "No B is C" empties B inside C.

Grading: The set of marked regions or cells must equal the key or an accepted alternative.

Verified: Automatic: each of the 8 regions tested against the condition.

S32-F1-Q5 · question page 757

LOGIC.SETS.02 · Harder categorical forms

"All A are B. Some A are C. So some B are C." This form is valid.

Answer: True

Reasoning: The A that is C is also B.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F2-Q1 · question page 758

LOGIC.SETS.02 · Existence and hidden assumptions

"All A are B. All A are C. So some B are C." This form is valid without any existence assumption.

Answer: False

Reasoning: If A is empty, both premises are true and B and C may not overlap.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S32-F2-Q2 · question page 759

LOGIC.SETS.02 · Existence and hidden assumptions

Which added premise makes "All A are B. All A are C. So some B are C." valid?

Answer: A. At least one A exists.

Reasoning: An A would be both B and C.

- ✗ At least one B exists.: B may exist outside C
- ✗ At least one C exists.: C may exist outside B
- ✗ All B are A.: still fails if A is empty

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S32-F2-Q3 · question page 760

LOGIC.SETS.02 · Existence and hidden assumptions

How many of these forms are valid (with no existence assumption)? (1) All A are B; all B are C; so all A are C. (2) No A is B; all C are A; so no C is B. (3) All A are B; some C are not B; so some C are not A. (4) Some A are B; all B are C; so some A are C.

Answer: 4

Reasoning: All four are valid: (1) nesting; (2) C inside A, A outside B; (3)

a C outside B cannot be in A; (4) the A that is B is also C.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S32-F2-Q4 · question page 761

LOGIC.SETS.02 · Existence and hidden assumptions

"All unicorns have one horn" is true if no unicorns exist.

Answer: True

Reasoning: An "all" statement about an empty set is vacuously true.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S32-F2-Q5 · question page 762

LOGIC.SETS.02 · Existence and hidden assumptions

"No square is a circle. Some circles are shaded. So some shaded things are not squares." Is it valid?

Answer: B. Yes: a shaded circle is not a square.

Reasoning: The shaded circle is a witness.

- ✗ No: shaded things may all be squares.: the shaded circles are not squares
- ✗ No: it needs squares to exist.: it does not
- ✗ Yes, but only if all circles are shaded.: not needed

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S33-M-Q1 · question page 764

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Put these four words in ladder order, starting with SAND, so that each word differs from the ne

Answer: SAND → SEND → SENT → RENT

Reasoning: SAND -> SEND (A to E) -> SENT (D to T) -> RENT (S to R).

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S33-M-Q2 · question page 765

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps in a ladder from BAND to BOAT?

Answer: 4

Reasoning: BAND -> BEND -> BENT -> BEAT -> BOAT (or through BEAD). Three steps are impossible: BAND has no neighbor on the list that is two steps from BOAT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-M-Q3 · question page 766

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST,

MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word fills the gap in the ladder SENT, ?, RANT?

Answer: B. RENT

Reasoning: SENT -> RENT -> RANT.

- ✗ BENT: BENT to RANT changes two letters
- ✗ SEND: SEND to RANT changes three letters
- ✗ PANT: SENT to PANT changes two letters

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S33-M-Q4 · question page 767

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is one step from both BEND and BEAT.

Answer: any values that satisfy: w == 'bent' or w == 'bead' (example: w = BENT)

Reasoning: BENT (BEND -> BENT, BEAT -> BENT) and BEAD (BEND -> BEAD, BEAT -> BEAD) both work.

Tempting answers the rule rejects: w = BELT; w = BOAT; w = BEST

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S33-M-Q5 · question page 768

LOGIC.WORD.02 · Find shortest one-letter-change paths through a supplied word list

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. A ladder from LAND to BELT can be done in 3 steps.

Answer: False

Reasoning: LAND and BELT differ in all 4 positions (L/B, A/E, N/L, D/T), and each step changes only one letter, so at least 4 steps are needed: LAND -> BAND -> BEND -> BENT -> BELT.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-B1-Q1 · question page 769

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word is one step from BEST?

Answer: A. REST

Reasoning: BEST -> REST changes only B to R.

- ✗ RENT: two letters differ from BEST
- ✗ POST: two letters differ
- ✗ BOAT: two letters differ

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S33-B1-Q2 · question page 770

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

In how many positions do the words MOST and PART differ?

Answer: 3

Reasoning: M/P, O/A, S/R differ; T matches.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-B1-Q3 · question page 771

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

A ladder from MAST to LAST can be done in one step.

Answer: True

Reasoning: Only M changes to L.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-B1-Q4 · question page 772

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from PORT to MOST?

Answer: 2

Reasoning: PORT -> POST -> MOST.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-B1-Q5 · question page 773

LOGIC.WORD.02 · Strategy: change the letters that must change, one at a time

If two words differ in 3 positions, every ladder between them has at least 3 steps.

Answer: True

Reasoning: Each step fixes at most one position.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S33-F1-Q1 · question page 774

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from BEND to PORT?

Answer: 6

Reasoning: BEND -> BENT -> RENT -> RANT -> PANT -> PART -> PORT. BEND and PORT differ in 4 positions, but the list forces a longer route.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F1-Q2 · question page 775

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many different shortest ladders go from BAND to BOAT?

Answer: 2

Reasoning: BAND-BEND-BENT-BEAT-BOAT and BAND-BEND-BEAD-BEAT-BOAT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F1-Q3 · question page 776

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Which word needs the MOST steps to reach from SAND (using shortest ladders)?

Answer: D. LAST

Reasoning: LAST is 11 steps from SAND: the chain RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST is the only way out to it, and LAST is at its far end.

- ✗ MOST: MOST takes 9 steps; LAST is two steps farther, through MAST
- ✗ BOLT: BOLT takes only 5 steps: SAND, BAND, BEND, BENT, BELT, BOLT
- ✗ PORT: PORT takes 7 steps; the chain continues past it to POST, MOST, MAST, LAST

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S33-F1-Q4 · question page 777

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Give a word from the list that is exactly one step from LAND and also one step from SEND.

Answer: any values that satisfy: w == 'sand' or w == 'lend' (example: w = SAND)

Reasoning: SAND (LAND -> SAND, SEND -> SAND) and LEND (LAND -> LEND, SEND -> LEND) both work.

Tempting answers the rule rejects: w = BAND; w = BEND

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S33-F1-Q5 · question page 778

LOGIC.WORD.02 · Longer ladders and detours

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. BOAT is one step from BOLT.

Answer: True

Reasoning: B, O and T stay in place; only the third letter changes (A to L).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-F2-Q1 · question page 779

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. What is the least number of steps from SAND to MOST?

Answer: 9

Reasoning: Breadth-first search over the list gives 9 steps: SAND, SEND, SENT, RENT, RANT, PANT, PART, PORT, POST, MOST.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F2-Q2 · question page 780

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Suppose PANT is removed from the list. What is the least number of steps from RANT to PART now?

Answer: 0

Reasoning: Without PANT, PART's only neighbor is PORT. The words PART, PORT, POST, MOST, MAST, LAST then link only to each other, and none of them is one step from RANT or any word connected to RANT, so no ladder exists.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F2-Q3 · question page 781

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. Every ladder from BAND to BOAT passes through BEND.

Answer: False

Reasoning: BAND -> SAND -> SEND -> SENT -> BENT -> BEAT -> BOAT avoids BEND. It is longer (6 steps) than the shortest ladders, but the statement is about every ladder.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S33-F2-Q4 · question page 782

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

Word list (the ONLY words allowed): SAND, BAND, BEND, SEND, SENT, BENT, BEAD, BEAT, BOAT, BELT, BOLT, BEST, REST, RENT, RANT, PANT, PART, PORT, POST, MOST, MAST, LAST, LAND, LEND. A ladder step changes exactly one letter and keeps the other letters in place; every word in the ladder must be on the list. How many words on the list are exactly one step from BENT?

Answer: 6

Reasoning: BEND, SENT, BEAT, BELT, BEST, RENT.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S33-F2-Q5 · question page 783

LOGIC.WORD.02 · Prove a ladder is shortest or impossible

A ladder must go from a 4-letter word to another 4-letter word that differs in all 4 positions. What is the least possible number of steps?

Answer: C. 4

Reasoning: Each of the 4 positions must change at least once.

- ✗ 3: each step changes only one position
- ✗ 2: each step changes only one position
- ✗ 8: that is not the least possible

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-M-Q1 · question page 785

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition says exactly the same thing as the clue "Exactly one locker lies between Cal and Dee"?

Answer: D. |C - D| = 2

Reasoning: One locker between them means their numbers differ by 2, in either order, so |C - D| = 2.

- ✗ |C - D| = 1: that says Cal and Dee are next to each other, with no locker between
- ✗ C - D = 2: that only allows Cal to the right of Dee; the clue allows either order
- ✗ |C - D| >= 2: that also allows two or more lockers between them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-M-Q2 · question page 786

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: Ava → Dee → Eli → Cal → Ben

Reasoning: Dee is in 2 or 4 (clue 5), and Cal is two away from Dee (clue 2). Testing each case with clues 1, 3 and 4 leaves only Ava 1, Dee 2, Eli 3, Cal 4, Ben 5.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 120 orderings of 5 items enumerated; 1 valid.

S34-M-Q3 · question page 787

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: True

Reasoning: Ava 1, Cal 2, Eli 3, Dee 4, Ben 5: clue 1 holds (1 < 5), clue 2 holds (|2 - 4| = 2), clue 3 holds, clue 4 fails (|5 - 2| = 3), clue 5 holds. Exactly one clue is broken.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S34-M-Q4 · question page 788

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: A. Clue 2

Reasoning: Cal is in 5 and Dee in 2: |5 - 2| = 3, so two lockers lie between them, not one.

- ✗ Clue 1: Ava 1 is left of Ben 4, so clue 1 holds
- ✗ Clue 4: Ben 4 and Cal 5 are next to each other, so clue 4 holds
- ✗ Clue 5: Dee is in locker 2, which is even, so clue 5 holds

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S34-M-Q5 · question page 789

LOGIC.CONSTRAINT.02 · Translate puzzle clues into exact conditions and test candidate arrangements against them

A row has nine lockers numbered 1 to 9. Eli's locker must meet every condition: it is odd-numbered; it is not at either end (not 1 and not 9); and it is at least 2 lockers away from locker 5 (|E - 5| >= 2). Give one locker number Eli could have.

Answer: any values that satisfy: isInteger(e) and isOdd(e) and e > 1 and e < 9 and abs(e - 5) >= 2 (example: e = 3)

Reasoning: Odd and not an end: 3, 5 or 7. At least 2 from 5 removes 5.

Eli could be in 3 or 7.

Tempting answers the rule rejects: e = 5; e = 9; e = 4; e = 1

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 4 listed mistakes fail it. Search of the stated domain: 2 valid answers.

S34-B1-Q1 · question page 790

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava is somewhere left of Ben"?

Answer: C. A < B

Reasoning: "Somewhere left of" only says Ava's number is smaller.

- ✗ A = B - 1: that says Ava is directly left of Ben; the clue allows a gap
- ✗ A > B: that puts Ava to the right of Ben
- ✗ A + 1 < B: that requires at least one locker between them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-B1-Q2 · question page 791

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ben and Cal are next to each other"?

Answer: B. |B - C| = 1

Reasoning: Next to each other means the numbers differ by 1, in either order.

- ✗ B - C = 1: that only allows Ben directly right of Cal
- ✗ |B - C| = 2: that puts one locker between them
- ✗ |B - C| <= 2: that also allows one locker between them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-B1-Q3 · question page 792

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. With five lockers, the clue "Eli is not at either end" means E is 2, 3 or 4.

Answer: True

Reasoning: The ends are lockers 1 and 5, so Eli is in 2, 3 or 4.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S34-B1-Q4 · question page 793

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The clue "Dee is directly left of Ava" means the same as D < A.

Answer: False

Reasoning: "Directly left" means the next locker: D = A - 1. D < A also allows a gap.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S34-B1-Q5 · question page 794

LOGIC.CONSTRAINT.02 · Strategy: write each clue as a condition on numbers

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Dee is in locker 2, and exactly one locker lies between Cal and Dee. Which locker is Cal in?

Answer: 4

Reasoning: |C - 2| = 2 gives C = 4 or C = 0, and there is no locker 0.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S34-F1-Q1 · question page 795

LOGIC.CONSTRAINT.02 · Harder clue words

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "At least two lockers lie between Ava and Eli"?

Answer: A. |A - E| >= 3

Reasoning: Two lockers between means the numbers differ by 3; "at least" allows more, in either order.

- ✗ |A - E| >= 2: that allows only one locker between them
- ✗ |A - E| = 3: that allows exactly two between them, not more
- ✗ A - E >= 3: that only allows Ava to the right of Eli

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-F1-Q2 · question page 796

LOGIC.CONSTRAINT.02 · Harder clue words

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Which condition matches the clue "Ben is in locker 3 unless Cal is"?

Answer: B. B = 3 or C = 3

Reasoning: If Cal is not in 3, Ben is. So one of them is in locker 3: B = 3 or C = 3.

- ✗ B = 3 and C is not 3: that rules out Cal in locker 3, but the clue allows it
- ✗ B is not 3 and C is not 3: that is the opposite: the clue needs one of them in locker 3
- ✗ C = 3: that rules out Ben in locker 3, but the clue allows it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-F1-Q3 · question page 797

LOGIC.CONSTRAINT.02 · Harder clue words

Six friends, Ava, Ben, Cal, Dee, Eli and Fay, each use one of six lockers numbered 1 to 6 from left to right. "Left of" means a lower number. Clues: (1) Ava is somewhere left of Ben. (2) Exactly two lockers lie between Cal and Dee. (3) Fay is somewhere right of Cal. (4) Dee is in an odd-numbered locker. (5) Cal is somewhere left of Ava. (6) Ben is not in locker 6. Put the friends in locker order,

Answer: Eli → Cal → Ava → Ben → Dee → Fay

Reasoning: Dee is odd and three away from Cal, so (Cal, Dee) is (2, 5), (4, 1) or (6, 3). Cal is left of Ava and Fay, so Cal is not 6. If Cal is 4, Ava and Fay need lockers 5 and 6, and Ben must be right of Ava, so Ava 5

and Ben 6, which clue 6 forbids. So Cal 2 and Dee 5. Ava, Ben and Fay take 3, 4 and 6 with Ava left of Ben and Ben not in 6: Ava 3, Ben 4, Fay 6, and Eli 1.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 720 orderings of 6 items enumerated; 1 valid.

S34-F1-Q4 · question page 798

LOGIC.CONSTRAINT.02 · Harder clue words

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. The two conditions A < B and B < A + 2 together say that Ben is directly right of Ava.

Answer: True

Reasoning: Locker numbers are whole numbers, so A < B < A + 2 forces B = A + 1.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S34-F1-Q5 · question page 799

LOGIC.CONSTRAINT.02 · Harder clue words

Seven lockers are numbered 1 to 7 from left to right. Give lockers for Ava and Ben that meet every clue: Ava is somewhere left of Ben; at least two lockers lie between them; Ben's locker is even-numbered.

Answer: any values that satisfy: isInteger(a) and isInteger(b) and a >= 1 and b <= 7 and a < b and b - a >= 3 and isEven(b) (example: a = 1, b = 4)

Reasoning: For example A = 1, B = 4. Other answers: (1, 6), (2, 6), (3, 6).

Tempting answers the rule rejects: a = 2, b = 4; a = 6, b = 2; a = 1, b = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 4 valid answers.

S34-F2-Q1 · question page 800

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: 4

Reasoning: Enumerating with clues 1, 2, 4 and 5 gives 4 arrangements, so clue 3 is needed for a single answer.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S34-F2-Q2 · question page 801

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. How many arrangements fit both clues "Eli is not at either end" and "Ben and Cal are next to each other" (and no other clues)?

Answer: 24

Reasoning: Treat Ben-and-Cal as a block. Eli is in 2, 3 or 4; for each, count the places for the block in either order and the arrangements of Ava and Dee in the last two lockers: 24 in all.

Grading: The game's answer checker compares the typed value with the key (equivalent

number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S34-F2-Q3 · question page 802

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Clues: (1) Ava is somewhere left of Ben. (2) Exactly one locker lies between Cal and Dee. (3) Eli is not at either end. (4) Ben and Cal are next to each other. (5) Dee is

Answer: True

Reasoning: The answer to the original puzzle has Dee in locker 2, and enumerating clues 1-4 with D = 2 leaves only Ava, Dee, Eli, Cal, Ben.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S34-F2-Q4 · question page 803

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. Which condition matches the clue "Ava and Ben are not both at the ends"?

Answer: D. At most one of Ava and Ben is in locker 1 or 5

Reasoning: "Not both" only rules out the case where both are at the ends; zero or one at an end is allowed.

- ✗ Neither Ava nor Ben is in locker 1 or 5: that forbids even one of them at an end
- ✗ Exactly one of Ava and Ben is in locker 1 or 5: that forbids both being away from the ends

✗ Ava and Ben are both in lockers 2, 3 or 4: that is the same as "neither is at an end"

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S34-F2-Q5 · question page 804

LOGIC.CONSTRAINT.02 · Test what each clue contributes

Five friends, Ava, Ben, Cal, Dee and Eli, each use one of five lockers numbered 1 to 5 from left to right, one friend per locker. "Left of" means a lower number. The letters A, B, C, D and E stand for the friends' locker numbers. In these puzzles, "X unless Y" means "if Y is false, then X is true." Under the clue "Ben is in locker 3 unless Cal is", an arrangement with Ava in locker 3 is allowed.

Answer: False

Reasoning: If Ava is in 3, Cal is not in 3, so the clue says Ben must be in 3. Ben cannot share the locker, so the arrangement breaks the clue.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S35-M-Q1 · question page 806

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement MUST be true?

Answer: C. Gears is on Monday or Tuesday.

Reasoning: Gears must be before Jazz, and Jazz is not on Friday, so Gears is not on Thursday or Friday. If Gears were on Wednesday, Jazz would be Thursday and Honey Friday, leaving Monday and Tuesday (consecutive) for Fossils and Kites. So Gears is on Monday or Tuesday.

✗ Honey is on Friday.: Fossils, Gears, Honey, Jazz, Kites (Monday to Friday) works with Honey on Wednesday

✗ Fossils is on Monday.: Gears, Kites, Honey, Jazz, Fossils works with Fossils on Friday

✗ Kites is on Friday.: Fossils, Gears, Jazz, Kites, Honey works with Kites on Thursday

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-M-Q2 · question page 807

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which statement CANNOT be true?

Answer: A. Jazz is on Tuesday.

Reasoning: If Jazz is on Tuesday, Gears is on Monday. Honey takes Wednesday or Friday, and Fossils and Kites take the other two of Wednesday, Thursday, Friday. If Honey is on Friday they are on Wednesday and Thursday (consecutive); if Honey is on Wednesday, they are on Thursday and Friday (consecutive). Either way condition 3 fails.

✗ Fossils is on Friday.: Gears, Kites, Honey, Jazz, Fossils meets every condition

✗ Kites is on Wednesday.: Fossils, Gears, Kites, Jazz, Honey meets every condition

✗ Honey is on Wednesday.: Fossils, Gears, Honey, Jazz, Kites meets every condition

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-M-Q3 · question page 808

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Fossils could be on Friday.

Answer: True

Reasoning: Gears, Kites, Honey, Jazz, Fossils (Monday to Friday) meets all four conditions.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S35-M-Q4 · question page 809

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many complete schedules meet all four conditions?

Answer: 10

Reasoning: Case by Honey's day. Honey Wednesday: 4 schedules. Honey Friday: 6 schedules. Total 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-M-Q5 · question page 810

LOGIC.CONSTRAINT.03 · Decide what must be true, what could be true and what cannot be true in a constrained arrangement

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Wednesday. Using Monday = 1, Tuesday = 2, Wednesday = 3, Thursday = 4 and Friday = 5, give a day number Ki

Answer: any values that satisfy: k == 1 or k == 2 or k == 5 (example: k = 5)

Reasoning: With Honey on Wednesday the schedules are F G H J K, G F H J K, G K H J F and K G H J F, so Kites could be on day 1, 2 or 5.

Tempting answers the rule rejects: k = 3; k = 4

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S35-B1-Q1 · question page 811

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. How many schedules meet both conditions?

Answer: 2

Reasoning: P on day 2: Q, P, R. P on day 3: Q, R, P.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-B1-Q2 · question page 812

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then Q must be on day 1.

Answer: True

Reasoning: The two schedules are Q, P, R and Q, R, P. Q is on day 1 in both.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S35-B1-Q3 · question page 813

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Then R must be on day 3.

Answer: False

Reasoning: In Q, R, P, R is on day 2. R could be on day 3 but need not be.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S35-B1-Q4 · question page 814

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

In a scheduling puzzle, what does it mean to say a statement CANNOT be true?

Answer: B. It is false in every schedule that meets the conditions.

Reasoning: Cannot be true means no valid schedule makes it true.

✗ It is false in at least one schedule that meets the conditions.: that means the statement need not be true; it could still be true in another

schedule

✗ It is true in exactly one schedule that meets the conditions.: then the statement could be true

✗ It is not mentioned in any of the conditions.: a statement can be forced false by conditions that never mention it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S35-B1-Q5 · question page 815

LOGIC.CONSTRAINT.03 · Strategy: list every case, then read must, could, cannot

Three talks, P, Q and R, are on days 1, 2 and 3, one per day. Conditions: P is not on day 1; Q is earlier than R. Which statement COULD be true?

Answer: C. P is on day 2.

Reasoning: Q, P, R meets both conditions and has P on day 2.

✗ P is on day 1.: the first condition forbids it

✗ R is on day 1.: Q must be earlier than R, so R is never first

✗ Q is on day 3.: Q must be earlier than R, so Q is never last

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S35-F1-Q1 · question page 816

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Monday, which statement MUST be true?

Answer: D. Gears is on Tuesday.

Reasoning: Fossils cannot be on Tuesday (next to Kites), Jazz cannot be

on Tuesday (Gears must come first), and Honey is on Wednesday or Friday, so Gears is on Tuesday.

- ✗ Honey is on Friday.: Kites, Gears, Honey, Jazz, Fossils puts Honey on Wednesday
- ✗ Fossils is on Wednesday.: Kites, Gears, Jazz, Fossils, Honey puts Fossils on Thursday
- ✗ Jazz is on Thursday.: Kites, Gears, Jazz, Fossils, Honey puts Jazz on Wednesday

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-F1-Q2 · question page 817

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Kites is on Friday, how many schedules are possible?

Answer: 2

Reasoning: Honey must be on Wednesday, and Fossils is not on Thursday (next to Kites). Fossils, Gears and Jazz fill Monday, Tuesday and Thursday with Gears before Jazz, so Jazz is on Thursday: F G H J K and G F H J K.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-F1-Q3 · question page 818

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from

Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Jazz is on Wednesday, then Honey must be on Friday.

Answer: True

Reasoning: Honey is on Wednesday or Friday, and Wednesday is taken by Jazz.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S35-F1-Q4 · question page 819

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Which ONE added condition would leave exactly one possible schedule?

Answer: A. Fossils is on Wednesday.

Reasoning: With Fossils on Wednesday, only Kites, Gears, Fossils, Jazz, Honey works.

- ✗ Fossils is on Friday.: two schedules remain: G K H J F and K G H J F
- ✗ Kites is on Thursday.: two schedules remain: F G J K H and G F J K H
- ✗ Honey is on Wednesday.: four schedules remain

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-F1-Q5 · question page 820

LOGIC.CONSTRAINT.03 · Must, could, cannot under an added condition

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose Honey is on Friday. Using Monday = 1 to Friday = 5, give the day numbers of Gears and Jazz in one schedule that meets

Answer: any values that satisfy: (g == 2 and (j == 3 or j == 4)) or (g == 1 and j == 3) (example: g = 2, j = 3)

Reasoning: Listing the schedules with Honey on Friday gives (Gears, Jazz) = (1, 3), (2, 3) or (2, 4). The tempting (1, 4) fails: Fossils and Kites would be on Tuesday and Wednesday, which are consecutive.

Tempting answers the rule rejects: g = 1, j = 4; g = 3, j = 4; g = 4, j = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S35-F2-Q1 · question page 821

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. How many of the five talks could be on Friday?

Answer: 3

Reasoning: Fossils, Honey and Kites each appear on Friday in some schedule. Jazz cannot (condition 4) and Gears cannot (it must be before Jazz).

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-F2-Q2 · question page 822

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Suppose condition 3 is replaced by "Fossils is on Monday." Which statement must be true?

Answer: C. Jazz is on Wednesday or Thursday.

Reasoning: The four schedules are F G H J K, F G J K H, F G K J H and F K G J H. Jazz is on Wednesday or Thursday in each.

- ✗ Gears is on Tuesday.: Fossils, Kites, Gears, Jazz, Honey has Gears on Wednesday
- ✗ Kites is on Friday.: Fossils, Gears, Jazz, Kites, Honey has Kites on Thursday
- ✗ Honey is on Friday.: Fossils, Gears, Honey, Jazz, Kites has Honey on Wednesday

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S35-F2-Q3 · question page 823

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Removing condition 4 adds exactly two new schedules.

Answer: False

Reasoning: Without condition 4 there are 14 schedules, which is 4 more than 10.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S35-F2-Q4 · question page 824

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. Add a fifth condition: no two of Gears, Honey and Jazz are on consecutive days. How many schedules remain?

Answer: 2

Reasoning: Gears, Jazz and Honey must be on days 1, 3 and 5, so Gears is Monday, Jazz Wednesday, Honey Friday. Fossils and Kites take Tuesday and Thursday in either order: 2 schedules.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S35-F2-Q5 · question page 825

LOGIC.CONSTRAINT.03 · Change the conditions and re-test

A club holds five talks, Fossils, Gears, Honey, Jazz and Kites, one on each day from Monday to Friday. Conditions: (1) Gears is earlier in the week than Jazz. (2) Honey is on Wednesday or Friday. (3) Fossils and Kites are not on consecutive days. (4) Jazz is not on Friday. If Gears is on Monday, then Fossils and Kites could be on Tuesday and Thursday.

Answer: True

Reasoning: Gears, Fossils, Jazz, Kites, Honey meets every condition.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-M-Q1 · question page 827

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

How many three-digit whole numbers (100 to 999) have digits that add up to 5?

Answer: 15

Reasoning: Case by the hundreds digit: 1 leaves 4 for two digits (5 ways), 2 leaves 3 (4 ways), 3 leaves 2 (3 ways), 4 leaves 1 (2 ways), 5 leaves 0 (1 way): 15.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-M-Q2 · question page 828

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 22?

Answer: 3

Reasoning: The minute digits add to at most 5 + 9 = 14, so the hour digits must add to at least 8: hours 8 and 9. Hour 8 needs minutes adding to 14: 59. Hour 9 needs 13: 49 and 58. Three times: 8:59, 9:49, 9:58.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-M-Q3 · question page 829

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

Find the two-digit whole number that equals 3 times the sum of its digits.

Answer: any values that satisfy: (isInteger(n) and n >= 10 and n <= 99 and n == 3 * (floor(n / 10) + mod(n, 10))) and isInteger(n) and n >= 0

(example: n = 27)

Reasoning: 27 = 3 x (2 + 7). Writing n = 10a + b, the condition 10a + b = 3a + 3b gives 7a = 2b, so a = 2, b = 7 is the only digit solution.

Tempting answers the rule rejects: n = 18; n = 72; n = 30

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S36-M-Q4 · question page 830

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

In how many ways can you make 30 cents using nickels (5 cents), dimes (10 cents) and quarters (25 cents)? You may use any number of each coin, including none; the order of the coins does not matter.

Answer: 5

Reasoning: With no quarter: 0, 1, 2 or 3 dimes, the rest nickels (4 ways). With one quarter: one nickel (1 way). Total 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-M-Q5 · question page 831

LOGIC.SEARCH.01 · Organize a complete case analysis and count or find every solution

Exactly 6 of the 24 orderings of the letters A, B, C and D have A before B and also C before D.

Answer: True

Reasoning: Each of the conditions "A before B" and "C before D" holds in half of the orderings, independently: 24 / 2 / 2 = 6. Listing them confirms

6.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-B1-Q1 · question page 832

LOGIC.SEARCH.01 · Strategy: make an organized list

How many two-digit numbers have digits that add up to 6?

Answer: 6

Reasoning: 15, 24, 33, 42, 51, 60: six.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-B1-Q2 · question page 833

LOGIC.SEARCH.01 · Strategy: make an organized list

In how many ways can you make 20 cents using only nickels and dimes? The order of the coins does not matter.

Answer: 3

Reasoning: 0 dimes and 4 nickels, 1 dime and 2 nickels, 2 dimes: 3 ways.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-B1-Q3 · question page 834

LOGIC.SEARCH.01 · Strategy: make an organized list

There are exactly 4 two-digit numbers whose digits multiply to 12.

Answer: True

Reasoning: Tens digit 2, 3, 4, 6 gives 26, 34, 43, 62. Other tens digits do not divide 12 into a single digit.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-B1-Q4 · question page 835

LOGIC.SEARCH.01 · Strategy: make an organized list

Ana lists the ways to make 15 cents from nickels and dimes: "3 nickels; 1 dime and 1 nickel." Is her list complete? The order of the coins does not matter.

Answer: D. Yes: she used 0 dimes and 1 dime, and 2 dimes is too much

Reasoning: Organize by the number of dimes: 0 dimes needs 3 nickels, 1 dime needs 1 nickel, 2 dimes is too much. Her two ways are all of them.

- ✗ No: 2 dimes and no nickels is missing: 2 dimes make 20 cents, not 15
- ✗ No: 1 dime alone is missing: 1 dime makes only 10 cents
- ✗ No: a nickel then a dime is a third way: the order of the coins does not matter

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S36-B1-Q5 · question page 836

LOGIC.SEARCH.01 · Strategy: make an organized list

There are 5 two-digit numbers whose digits add up to 3.

Answer: False

Reasoning: By tens digit: 12, 21, 30. There are 3.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-F1-Q1 · question page 837

LOGIC.SEARCH.01 · Counting with extra conditions

How many three-digit numbers have three different digits that add up to 6?

Answer: 14

Reasoning: The digit sets are {0, 1, 5}, {0, 2, 4} and {1, 2, 3}. Each set with a 0 gives 4 numbers (0 cannot lead), and {1, 2, 3} gives 6: 4 + 4 + 6 = 14.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F1-Q2 · question page 838

LOGIC.SEARCH.01 · Counting with extra conditions

In how many ways can 10 be written as a sum of 1s and 2s if the order matters? (For example, 3 can be written 3 ways: 1 + 1 + 1, 1 + 2 and 2 + 1.)

Answer: 89

Reasoning: Organize by the first number: ways(n) = ways(n - 1) + ways(n - 2). Starting 1, 2, 3, 5, 8, ... gives 89 for 10.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F1-Q3 · question page 839

LOGIC.SEARCH.01 · Counting with extra conditions

Find a three-digit number whose three digits are all odd, all different, and add up to 15.

Answer: any values that satisfy: isInteger(n) and n >= 100 and n <= 999 and isOdd(floor(n / 100)) and isOdd(mod(floor(n / 10), 10)) and isOdd(mod(n, 10)) and floor(n / 100) + mod(floor(n / 10), 10) + mod(n, 10) == 15 and floor(n / 100) != mod(floor(n / 10), 10) and floor(n / 100) != mod(n, 10) and mod(floor(n / 10), 10) != mod(n, 10) (example: n = 159)

Reasoning: The digit sets are {1, 5, 9} and {3, 5, 7}; any order works, for example 159 or 357.

Tempting answers the rule rejects: n = 555; n = 168; n = 339

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 12 valid answers.

S36-F1-Q4 · question page 840

LOGIC.SEARCH.01 · Counting with extra conditions

There are more ways to make 50 cents from nickels and dimes than from dimes and quarters. The order of the coins does not matter.

Answer: True

Reasoning: Nickels and dimes: 0 to 5 dimes, 6 ways. Dimes and quarters: 5 dimes, or 2 quarters, 2 ways (1 quarter leaves 25 cents, which dimes cannot make).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-F1-Q5 · question page 841

LOGIC.SEARCH.01 · Counting with extra conditions

How many rectangles of any size (including squares) can be traced along the lines of a grid that is 2 squares tall and 3 squares wide?

Answer: B. 18

Reasoning: By size: 1 by 1: 6; 1 tall by 2 wide: 4; 1 by 3: 2; 2 tall by 1: 3; 2 by 2: 2; 2 by 3: 1. Total 18.

- ✗ 6: that counts only the single squares
- ✗ 15: that leaves out the three rectangles 2 squares tall and 1 wide
- ✗ 17: that leaves out the whole grid

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S36-F2-Q1 · question page 842

LOGIC.SEARCH.01 · Searches with many cases

A 12-hour digital clock shows times from 1:00 to 12:59: the hour (1 to 12, no leading zero) and then two minute digits (00 to 59). At how many times do the shown digits add up to exactly 6?

Answer: 36

Reasoning: Case by the hour: its digit sum s leaves 6 - s for the minutes; count minute pairs (tens 0 to 5) for each hour and add.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F2-Q2 · question page 843

LOGIC.SEARCH.01 · Searches with many cases

In how many ways can you make 1 dollar (100 cents) using nickels, dimes and quarters? The order of the coins does not matter.

Answer: 29

Reasoning: Case by quarters. 0 quarters: 0 to 10 dimes (11 ways). 1 quarter leaves 75: 0 to 7 dimes (8). 2 quarters leave 50: 6. 3 quarters leave 25: 3. 4 quarters: 1. Total 29.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F2-Q3 · question page 844

LOGIC.SEARCH.01 · Searches with many cases

How many positive whole numbers less than 1000 have digits that add up to 4?

Answer: 15

Reasoning: One digit: 4 (1). Two digits: 13, 22, 31, 40 (4). Three digits: leading digit 1 to 4 leaves 3, 2, 1, 0 for two digits: 4 + 3 + 2 + 1 = 10. Total 15.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S36-F2-Q4 · question page 845

LOGIC.SEARCH.01 · Searches with many cases

Every whole number of cents from 8 to 30 can be paid exactly using only 3-cent and 5-cent stamps.

Answer: True

Reasoning: 8 = 3 + 5, 9 = 3 + 3 + 3, 10 = 5 + 5. Once three amounts in a row work, adding a 3-cent stamp reaches every larger amount.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.